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ORIGINAL ARTICLE

Short-term load forecasting: cascade intuitionistic fuzzy time series—univariate and bivariate models

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Abstract

Short-term load forecasting (STLF) is essential for developing reliable and sustainable economic and operational strategies for power systems. This study presents a forecasting model combining cascade forward neural network (CFNN) and intuitionistic fuzzy time series (IFTS) models for STLF. The proposed cascading intuitionistic fuzzy time series forecasting model (C-IFTS-FM) offers the advantage of CFNN using the links of both linear and nonlinear to model fuzzy relations between inputs and outputs. Moreover, it offers a more reliable and realistic approach to uncertainty, taking notice of also the degree of hesitation. C-IFTS-FM works in univariate structure when it uses only hourly load data, and in bivariate structure when it uses hourly load data and hourly temperature time series together. The conversion of time series into IFTS is realized with intuitionistic fuzzy c-means (IFCM). Thus, the membership and non-membership values for each data point are produced. In modelling process, membership and non-membership values, in addition to actual lagged observations, are used as input of the CFNNs. The effectiveness of C-IFTS-FM on test sets for both structures was discussed comparatively via different error criteria, in addition, the convergence time was examined, and also the fit of forecasts and observations was presented with different illustrations. Among different combinations of hyperparameters, in the best case, approximately 86% better accuracy is achieved than the best of the others, while even in the case of the worst of hyperparameters combination, the accuracy was improved by over 20% for the PSJM data sets. For HEXING, CHENGNAN, and EUNITE data sets, these progress rates reached approximately 90% in the best case.

Keywords Short-term load forecasting · Cascade forward neural network · Intuitionistic fuzzy time series · Univariate and bivariate time series forecasting

Abbreviations

STLF	Short-term load forecasting
CFNN	Cascade forward neural network
IFTS	Intuitionistic fuzzy time series
C-IFTS-FM	Cascading intuitionistic fuzzy time series forecasting model
IFCM	Intuitionistic fuzzy C-means
ARIMA	Auto-regressive integrated moving average
SARIMA	Seasonal auto-regressive integrated moving average
ANN	Artificial neural network
BNN	Bayesian neural network
LSTM	Long short-term memory
FTS	Fuzzy time series
CNN	Convolutional neural network
LightGBM	Light gradient-boosting machine
FB-Prophet	Facebook prophet
IFSs	Intuitionistic fuzzy sets
UC-IFTS-FM	Univariate cascading intuitionistic fuzzy time series forecasting model
BC-IFTS-FM	Bivariate cascading intuitionistic fuzzy time series forecasting model
CFSs	Classical fuzzy sets
IFE	Intuitionistic fuzzy entropy
HLU	Hidden layer unit

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OLU	Output layer unit
ILU	Input layer unit
PSJM	Power Supply Johor in Malaysia
T-MEP	Türkiye's monthly electricity production
EXIST	Energy exchange Istanbul
TP	Total production
NG	Electricity is produced from natural gas
HYD	Electricity is produced from hydro
COAL	Electricity is produced from coal
WIND	Electricity is produced from wind
FO	Electricity is produced from fuel oil
BIO	Electricity is produced from bioenergy
GEO	Electricity is produced from geothermal
MAPE	Mean absolute percentage error
RMSE	Root mean square error
MdRAE	Median relative absolute error
PWFTS	Probabilistic weighted fuzzy time series
WFTS	Weighted fuzzy time series
IWFTS	Integrated weighted fuzzy time series
FTS-CNN	Combined method for convolutional neural network and fuzzy time series
CI	Confidence interval
GEP	Gene expression programming
GEP-LS	Gene expression programming based on least square
GEP-KLS	Gene expression programming based on K-means and least square
QCLF	Quantitative combination load forecasting
PLLR	Pattern-based local linear regression
BPNN	Back-propagation neural network
HW Add.	Holt–Winters exponential smoothing—additive
HW Mult.	Holt–Winters exponential smoothing—multiplicative
ETS	State space models (error–trend–seasonality)
TBATS	Trigonometric seasonality, box–cox transformation, ARMA errors, trend components, seasonal components models
STL-SARIMA	Hybrid model of seasonal trend loess decomposition and SARIMA
STL-HW Add.	Hybrid model of seasonal trend loess decomposition and Holt–Winters exponential smoothing—additive
STL-HW Mult.	Hybrid model of seasonal trend loess decomposition and Holt–Winters exponential smoothing—multiplicative
STL-ETS	Hybrid model of seasonal trend loess decomposition and state space models (error–trend–seasonality)
ANN	Artificial neural network
STL-ANN	Hybrid model of seasonal trend loess decomposition and artificial neural network
STL-LSTM	Hybrid model of seasonal trend loess decomposition and long short-term memory
STL-CNN	Hybrid model of seasonal trend loess decomposition and convolutional neural network
MV-ANN	Multivariate artificial neural network model
MV-LSTM	Multivariate long short-term memory model
MV-CNN	Multivariate convolutional neural network model
ARDL	Auto-regressive distributed lag model

1 Introduction

1.1 General information and literature review

Load forecasting is an issue having a vital effect on making today's and future plans more sustainable and reliable to render the safe and efficient operation of power systems. Considering the variety of external impact factors, high levels of uncertainties that belong to these factors are obviously seen. So, obtaining an accurate load estimation is directly associated with power system management, hence the economy, and has a considerable effect on them. The more forecasting errors are reduced, the lower the operating cost will be. On the other hand, inaccurate forecasting results may cause huge losses for electric power companies by resulting in extra cost production and insufficient electricity supply. Therefore, an accurate and reliable STLF is a significant achievement for most energy companies. This goal motivates scientists and researchers to develop models to obtain accurate and practical methods. Over the years, to be able to improve the forecasting performance for the load demands, different methods have been developed. These are referred to under two main groups. The first one, probabilistic methods, includes statistical-based approaches. The second, non-probabilistic methods, is classified as computation-based and fuzzy-based methods [1].

As statistical-based models, linear regression, Kalman filtering, auto-regressive integrated moving average models (ARIMA), seasonal auto-regressive integrated moving average models (SARIMA), and stochastic process models were the most preferred models for the short-term load forecasting literature. In the early stage of load forecasting, to be able to describe the developed forecasting system for the Imatra Power Company, Vähäkyla et al. [2] proposed a forecasting model which uses the ARIMA model for STLF. While Tafreshi and Farhadi [3] performed the linear regression method, Kalman filtering was used in Zheng et al. [4] study. Auto-regressive integrated moving average was used by Hippert et al. [5]. On the other hand, Weron [6] and Taylor and McSharry [7] proposed studies based on statistical forecasting models for the load forecasting problems.

But just like the other statistical-based methods, these models have some deficiencies. Although they are effective in dealing with linear structures, some strict assumptions need to be satisfied to be able to work with these models. Moreover, when dealing with nonlinear or complex structures, again statistical-based methods are not effective. In these conditions, to handle some of these problems artificial neural network-based models have been started and frequently used for the STLF see, for example, [8–12]. Artificial neural networks (ANNs) have become drastically popular models thanks to some properties such as fault tolerance, powerful capability in pattern recognition, and distributed associative memory[8]. With the aim of modelling electricity demand for Canada, Zahidi et al. [13] used an adaptive neural network by filtering the inputs with Pearson correlation. Ghofrani et al. [14] presented a hybrid model which includes Bayesian neural network (BNN) and a wavelet transform, in order to obtain load characteristics for BNN training with detail.

Apart from ANN-based models, deep learning models were also the preferred methods for the load forecasting studies. Khan et al. [15] proposed a hybrid method that combines the random forest and recursive feature elimination methods with a deep neural network to forecast loads for a week. To be able to forecast the effect of the power load consumption Atef and Eltawil preferred to use long short-term memory (LSTM) and bidirectional LSTM networks [16]. Kumar et al. [17] introduced a forecasting model in order to handle the nonlinearity and seasonality problem in power load data forecasting. It is also obvious that all these models may fail to model the data accurately if the data sets contain ambiguous and linguistic terms. Given the fact that almost all of the daily lifetime series data sets contain vagueness, the interest in fuzzy time series methods for the STLF is also inevitable. Fuzzy time series (FTS) models, with their distinguished advantages, have started to be preferred in short-term forecasts as well as in many areas. A new weighted FTS model was designed by Yu [18] to be able to forecast the stock market. Also, for stock price forecasting, Cagcag Yolcu and Alpaslan [19], Trierweiler Ribeiro et al. [20], and Ren et al. [21] proposed hybrid models. A hybrid model which combines harmony search algorithm with weighted FTS was introduced by Sadaei et al. [22] for load data sets. On the other hand, in a different study, a combined seasonal auto-regressive fractional moving average model with FTS was proposed by Sadaei et al. [23] for STLF. Moreover, in some other studies, FTS models have been used as a forecasting tool for STLF [24–26]. Apart from these, Sadaei et al. presented a hybrid study which integrated the convolutional neural network (CNN) and FTS for short-term load forecasting [27]. Wang et al. recommended using a deep autoencoder for STLF [28], while Zhao et al. used a global prediction method based on pre-trained autoencoder and deep LSTM model [29]. Berahmand et al., in their studies, discussed autoencoders and their applications in machine learning in detail [30, 31]. In electric power system sector, a hybrid wavelet stacking ensemble model was proposed to forecast insulators contamination [32]. Shakeel et al. developed a hybrid model for district heating load forecasting based on the LightGBM and FB-Prophet [33]. A novel hybrid model based on multivariate empirical mode decomposition and support vector regression with hyperparameters optimized by particle swarm optimization was introduced for heat load [34]. Additionally, some studies emphasize that accurate short-term load forecasts should include temperature information to ensure the reliable and efficient operation of power systems [35–37]. In this context, this study introduces not only a univariate forecasting model but also a bivariate model that includes hourly temperature time series as an explanatory variable.

Considering that all these fuzzy-based and computational-based models are investigated in detail, some significant deficiencies can be mentioned. First of all, computation-based models do not offer an approach to the uncertainty contained in time series in forecasting problems such as STLF and focus only on prediction accuracy. However, it is inevitable that time series, including energy data sets, contain uncertainty by their nature. To address uncertainty, more complex and sophisticated approaches, such as models incorporating fuzzy logic, are required. In this context, while fuzzy-based and hybrid models have been proposed for STLF [24, 25, 27], they also have fundamental problems. The first problem is that in almost all of these studies, fuzzy set indices are taken into account instead of membership values, and relationships are created on these, which causes a loss of information that will negatively affect forecasting performance. Secondarily, in all these fuzzy-based models, the neutrality degree of time series has not been taken into account. This means that characterizing the uncertainty of the data which belongs to non-membership degrees and the neutrality degrees of time series has not been considered in a comprehensive manner. Also, in the decision-making process, hesitations always exist. Since there are always some factors that affect the decision process, in this study, a model based on intuitionistic fuzzy sets introduced by Atanassov [38], which incorporates all relevant information, is proposed. In this way, using information about both membership and non-membership degrees will provide more realistic and powerful forecasting results. With the introduction of IFSs, these problems have been relatively eliminated. There is no introduced study that uses IFSs in the STLF literature. Furthermore, while some studies on STLF propose linear models [3, 4], others identify relationships using nonlinear models [8, 9, 11, 27]. However, data sets can contain both linear and nonlinear relationships, together. This study addresses this issue by employing a cascaded forward neural network to uncover and identify the relationships.

In this study, we aim to achieve more effective forecasting results in the STLF problem by incorporating both the hesitation degree and the inherent uncertainty in the data through an intuitionistic fuzzy sets-based approach. Considering the STLF literature, this study is notable for being the first to utilize IFSs in the field of STLF. On the other hand, to be able to reveal both and identify linear and nonlinear relationships between inputs and outputs, we leverage a cascade neural network with such capability. Therefore, we called the proposed forecasting model the cascade intuitionistic fuzzy time series forecasting model. C-IFTS-FM is capable of operating in both univariate and bivariate structures. In a univariate structure, C-IFTS-FM uses only hourly load data to model the load and to obtain the short-term forecasts of load (UC-IFTS-FM). In the bivariate structure, C-IFTS-FM also uses hourly temperature time series besides the hourly load data (BC-IFTS-FM). Additionally, Türkiye's monthly electricity production data from January 2012 to May 2022 were analysed using the proposed univariate model to demonstrate its performance.

C-IFTS-FM converts these time series with the real observations into IFTS by performing the IFCM algorithm. Besides the real observations of hourly load and temperature time series, both the non-membership and the membership values obtained in this way are also used in the forecasting process.

1.2 Motivation and contributions

STLF plays a significant role for the power system in terms of determining economic formulations, providing reliable and secure operating strategies. The main purpose of the STLF function can be summarized with titles Gross and Galiana [39].

- To provide load estimates for basic production scheduling functions
- To evaluate the power system security at any time
- To provide information relevant to timely dispatch

Apart from these important purposes, STLF has a huge usage area. STLF applications are crucial for hydro scheduling, as they help determine optimal reservoir releases and powerhouse generation levels. Moreover, STLF is necessary to determine hourly minimum-cost strategies for commissioning and shutting down units in thermal systems. Also, STLF is used to schedule the hourly operation of various resources and minimize production costs for mixed hydro and thermal systems. On the other hand, STLF plays an active role in ensuring power system security. System load forecasting is essential for detecting future conditions where the power system might be vulnerable. One of the other application areas of STLF is also providing system dispatchers with timely information, which helps them to operate the system economically and reliably by taking advantage of the latest weather forecasting and random behaviour.

When evaluating all these areas of use, the significance of STLF becomes evident. Therefore, developing an STLF model that offers high reliability, accuracy, and validity is crucial. However, obtaining accurate load forecasts is challenging, as various external factors influence loads. In this study, we aim to address this challenge by proposing a model that combines intuitionistic fuzzy clustering with cascade neural networks to achieve high accuracy. Forecasting tools based on artificial intelligence and fuzzy logic have been widely used to support smart energy management in recent years. The proposed model leverages the fundamental advantages of neural networks and the uncertainty-handling capabilities of IFSs. Intuitionistic fuzzy sets incorporate membership degrees, non-membership degrees, and hesitation degrees, providing a richer and more comprehensive representation of uncertainty. These additional components of information can result in more accurate modelling of complex systems. Studies have shown that incorporating both membership and non-membership degrees as inputs in the models can enhance forecasting performance. By leveraging detailed information about the degree of membership and non-membership of observations, proposed models can make more informed forecasts. Moreover, what makes the proposed forecasting model superior and distinct from other models is its ability to simultaneously model both linear and nonlinear relationships between inputs and outputs, thanks to its cascaded analysis structure. The C-IFTS-FM offers several contributions, fundamental features, and advantages compared to current models aimed at STLF.

C-IFTS-FM offers a more realistic approach by modelling the uncertainty included in the time series, taking into account the degree of hesitation (to compare fuzzy-based models).

C-IFTS-FM uses both membership and non-membership values determined for time series (to compare fuzzy-based models).
C-IFTS-FM does not need to convert the data to an image and therefore does not include any loss of information due to conversion (to compare CNN-based models).
C-IFTS-FM has the ability of modelling both types of relationships between inputs and outputs (linear and nonlinear), thanks to its cascading structure (when compared with fuzzy-based and neural network-based models).
C-IFTS-FM is capable of operating in both univariate and bivariate structures, and thus, it can turn into a more effective forecasting tool by taking into account other variables that may have an effect on the load.

With these features, the proposed C-IFTS-FM shows superior performance in terms of different evaluation metrics and aspects as well for STLF.

2 Background

The background on which the proposed forecasting models are based can be given by mentioning some points. To determine the membership and non-membership degrees, the used clustering algorithm is presented in general terms. Moreover, in this section, brief information is given about the cascaded neural networks that enable the proposed forecasting models to determine linear and nonlinear relationships together, that is, their cascading structure is explained.

2.1 IFCM

IFSs establish a mathematical framework based on FSs to bring out and characterize vagueness and uncertainty existing inner of the data. IFSs have some distinct advantages over classical fuzzy sets (CFSs). CFSs [40] only take notice of the membership function, $\mu(x)$. However, IFSs deal with non-membership $v(x)$ and membership functions together.

When X is a universe of discourse, an IFS can be defined as in Eq. (1)

$$A = \{x, \mu_A(x), v_A(x)/x \in X\} \tag{1}$$

Here, for an element $x \epsilon X$, while $\mu_A(x) \rightarrow \lceil 0,1 \rceil$ depicts the membership degree, $\mu_A(x) \rightarrow \lceil 0,1 \rceil$ represents the non-membership degree under the condition given by Eq. (2).

$$0 \leq \mu_A(x) + v_A(x) \leq 1 \tag{2}$$

As another specific measure of uncertainty, $\pi_A(x)$, the degree of hesitation represents the lack of knowledge in defining membership and non-membership degrees and is defined as:

$$\pi_A(x) = 1 - \mu_A(x) - v_A(x) \tag{3}$$

It is clear from Eq. (3) that $0 \leq \pi_A(x) \leq 1$. For all $x \epsilon X$, if $\mu_A(x) = v_A(x) = 0$, the IFS is said to be purely intuitionistic. On the other hand, for all $x \epsilon X$, if $\pi_A(x) = 0$, the IFS transforms into a CFS.

The objective function to be optimized in IFCM [41], given in Eq. 4, consists of two components. The first is the modified objective function of FCM using IFS. The second is the intuitionistic fuzzy entropy (IFE).

$$J_{\mathrm{IFCM}} = \sum\nolimits_{i=1}^{c} \sum\nolimits_{k=1}^{n} u_{ik}^{*m} d_{ik}^{2} + \sum\nolimits_{i=1}^{c} \pi_i^* exp\left(1 - \pi_i^*\right) \tag{4}$$

Here, c and n represent the number of clusters and the number of data set points, respectively. In J_{IFCM}, u_{ik}^* depicts the intuitionistic fuzzy membership and calculated by Eq. (5)

$$u_{ik}^* = u_{ik} + \pi_{ik} \tag{5}$$

Also, u_{ik} is equivalent to the membership for CFS of the k th observation set in i th set. Consequently, the degree of hesitation, π_{ik}, can be given as follows:

$$\pi_{ik} = 1 - u_{ik} - \left(1 - u_{ik}^{\alpha}\right)^{\frac{1}{\alpha}}, \alpha > 0 \tag{6}$$

and is obtained via an intuitionistic fuzzy complement of Yager:

$$N(X) = (1 - x^{\alpha})^{\frac{1}{\alpha}}, \alpha > 0 \tag{7}$$

The alpha-cut (α-cut) determines the set of elements whose membership degree in A is at least α. Herewith, the IFS becomes

$$A_{\lambda}^{\mathrm{IFS}} = \left\{x, \mu_A(x), (1 - \mu_A(x)^{\alpha})^{\frac{1}{\alpha}}/x \in X\right\} \tag{8}$$

and

$$\pi_i^* = \frac{1}{N} \sum\nolimits_{k=1}^{N} \pi_{ik}, k \in [1, N] \tag{9}$$

IFE, the second component of the objective function, indicates the amount of fuzziness or uncertainty in a set. Thus, IFE as a measure of the degree of intuitionism can be defined as in Eq. (10).

$$\mathrm{IFE}(A) = \sum_{i=1}^{N} \pi_A(x_i)\exp(1 - \pi_A(x_i)), k \in [1, N] \quad (10)$$

also, the degree of hesitation

$$\pi_A(x_i) = 1 - \mu_A(x_i) - \nu_A(x_i) \quad (11)$$

The cluster centres can be modified as in Eq. (12)

$$v_i^* = \frac{\sum_{k=1}^{n} u_{ik}^* x_k}{\sum_{k=1}^{n} u_{ik}^*} \quad (12)$$

The maximum number of iterations can be taken as the stopping criterion for the process, or it can be adjusted according to the difference of the membership degrees in successive iterations. In this case, the process is stopped when the maximum number of iterations is reached or when $max_{ik}\left|u_{ik}^{*new} - u_{ik}^{*previous}\right| \leq \varepsilon$ (ε is pre-defined value).

2.2 Cascade forward neural network

Cascade forward neural network was proposed by Demuth [42] inspired by the cascade correlation approaches of Fahlman [43], and Fahlman and Lebiere [44]. Like other classical feed-forward multilayer neural networks, the architecture of CFNN consists of input, output, and hidden layer(s) [45]. However, the main feature that distinguishes this network from existing networks is that each neuron layer is associated with all previous neuron layers. However, the main distinguishing aspect of this network from existing networks is that it accommodates the nonlinear relationship between input and output by not eliminating the linear relationship between the two. The sigmoid activation function used in the hidden layer and the linear activation function used in the output layer provide this feature. A simple CFNN architecture with two hidden layers is given in Fig. 1.

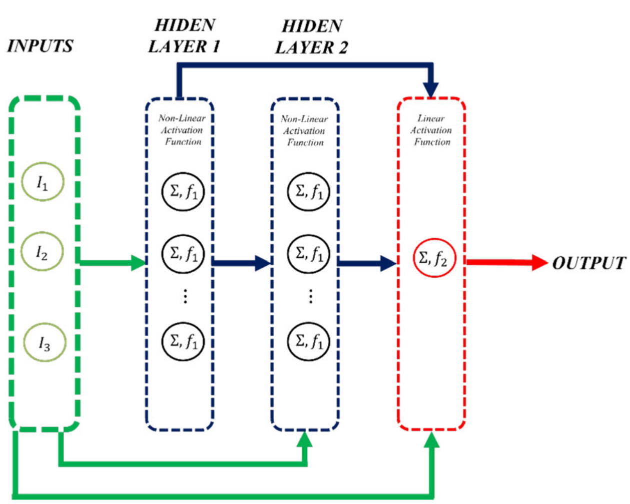


Fig. 1 The architecture of C-IFTS-M designed for STLF

3 The proposed methodology

3.1 C-IFTS-FM

Today, considering the increasing population and decreasing energy resources, national and global energy management is really important in terms of the sustainability of life. Load forecasting is one of the basic and most important elements of energy management. At this point, various forecasting tools act as a pioneer and guide for load forecasting as in many other areas. Especially in recent years, forecasting tools based on artificial intelligence and fuzzy logic are widely used to provide a service for smart energy management. In this study, a new forecasting model/tool is proposed for short-term load forecasting. The introduced model uses the basic advantages of neural networks and the uncertainty approach of IFSs. Moreover, the aspect that makes the proposed forecasting model superior and distinguishes it from other models is the ability of accommodating the nonlinear relationship between input and output by not eliminating the linear relationship between the two, thanks to its cascade analysis structure. The C-IFTS-FM is based on two definitions given in the following text and introduced in this study.

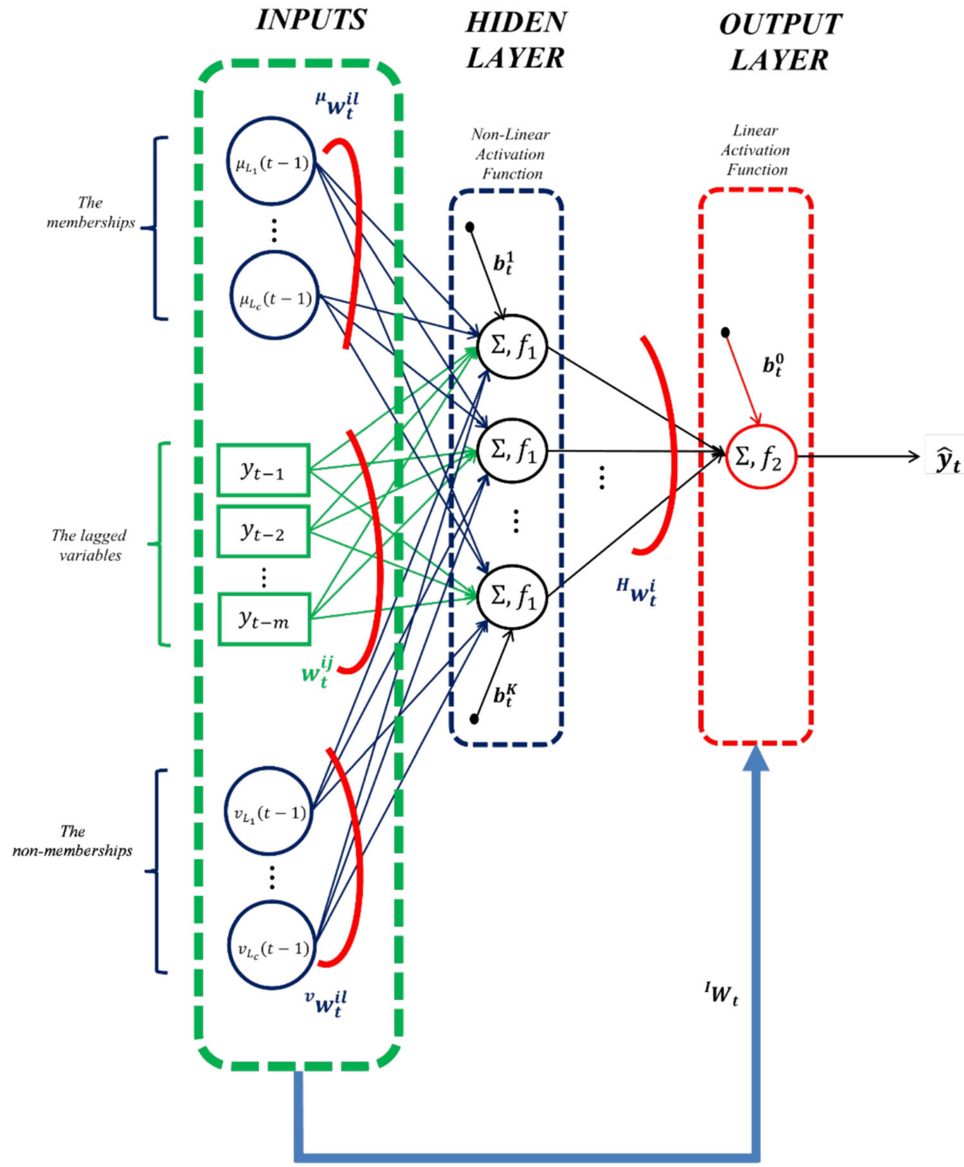
3.2 UC-IFTS-FM

C-IFTS-FM can be considered a model combining intuitionistic fuzzy clustering and cascade neural networks. By using IFCM clustering algorithm, the memberships and non-memberships values are determined for time series observations. And these membership and non-membership values are used as inputs of cascade neural network besides the lagged observations of crisp time series. The target values are also composed of crisp time series observation at t time. The structure of UC-IFTS-M can be given in Fig. 2.

UC-IFTS-FM, to determine the intuitionistic relationships between inputs and outputs, uses a cascade neural network composed of three layers as input, hidden, and output layers. A linear activation function (f_2) is executed in the output layer, while a nonlinear activation function (f_1) is used in hidden layer units (HLUs). The unit in the output layer (OLU) is fed directly with the inputs as well as with the outputs generated by processing the inputs in the hidden layer with a nonlinear activation function. Feeding the output layer with direct inputs models the effect of linear relationships, while feeding with the hidden layer output determines the effect of nonlinear relationships on the output.

Definition 5 (*UC-IFTS-FM*) Let IF_t be an IFTS, $I_1, I_2, \cdots, I_c$ are IFSs on the universal set, and $\mu_{I_l}(t)$, $\nu_{I_l}(t)$ are the degree of membership and non-membership values

Fig. 2 The structure of UC-IFTS-M designed for STLF



of the tth observation in the lth IFS, respectively. The high-order UC-IFTS-FM can be defined as in Eq. (13).

In Eq. (13), while F and F1 symbolize the linear functions, F2 symbolizes a nonlinear function obtained from an estimation tool. In this step an artificial neural network has been used to reveal the nonlinear relationships but different estimation tools can be chosen as well. ε_t represents the error term. Thanks to the cascaded structure of the model, both linear and nonlinear relationships can be modelled synchronously at the same time.

$$\begin{aligned} Y_t = {} & F\big(F_1\big(Y_{t-1}Y_{t-2}\cdots, Y_{t-m}, \mu_{I_1}(t-1), \\ & \mu_{I_2}(t-1), \cdots, \mu_{I_c}(t-1), v_{I_1}(t-1), v_{I_2}(t-1), \cdots, \\ & v_{I_c}(t-1)\big), F_2\big(Y_{t-1}Y_{t-2}\cdots, Y_{t-m}, \mu_{I_1}(t-1), \\ & \mu_{I_2}(t-1), \cdots, \mu_{I_c}(t-1), v_{I_1}(t-1), v_{I_2}(t-1), \cdots, \\ & v_{I_c}(t-1)\big)\big) + \varepsilon_t \end{aligned} \tag{13}$$

where $Y_{t-1}Y_{t-2}\cdots, Y_{t-m}$ are mth order lagged real valued time series for tth time point, $\mu_{I_1}(t-1), \mu_{I_2}(t-1), \cdots, \mu_{I_c}(t-1)$ are the first-order lagged degrees of membership of the tth observation in each intuitionistic fuzzy set also

$v_{I_1}(t-1), v_{I_2}(t-1), \cdots, v_{I_c}(t-1)$ are the first-order lagged degrees of membership of the tth observation in each intuitionistic fuzzy set.

In the structure given by Fig. 2;

$\mu_{Lc}(t-1)$: The membership input, associated with cth IFS, for the data point at the $t-1$ time.
$v_{Lc}(t-1)$: The non-membership input, associated with cth IFS, for the data point at the $t-1$ time.
y_{t-j}: Real input related to jth lagged variable.
$\widehat{y}_t$: The output for the t time.
μw_t^{il}: For membership component, the weight between lth input layer unit (ILU) and ith HLU$\left(i=\overline{1,K}; l=\overline{1,c}\right)$.
${}_v w_t^{il}$: For non-membership component, the weight between lth ILU and ith HLU$\left(i=\overline{1,K}; l=\overline{1,c}\right)$.
w_t^{ij}: For lagged variables with real values, the weights between jth ILU and ith HLU$\left(i=\overline{1,K}; j=\overline{1,m}\right)$.
b_t^i: The bias for ith HLU$\left(i=\overline{1,K}\right)$.
Hw_t^i: The weight between ith HLU and the OLU$\left(i=\overline{1,K}\right)$.
IW_t: The vector of weights between ILUs and the H$\left(IW_t=[Iw_t^1 Iw_t^2 \cdots Iw_t^{2\times l+m}]\right)$.
b_t^o: The bias of OLU.

The outputs of HLU are obtained by Eq. (14).

$$o_i = f_1\left(\left(\sum_{l=1}^{c}\mu w_t^{il}\mu_{Ll}(t-1)\right)+\left(\sum_{l=1}^{c} v w_t^{il} v_{Ll}(t-1)\right) + \left(\sum_{j=1}^{m} w_t^{ij} y_{t-j}\right)+b_t^i\right), i=1,2,\ldots,K \tag{14}$$

Here f_1 represents the sigmoid activation function and $f_1(x)=\frac{1}{1+exp(-x)}$. The outputs of the system are obtained, by using the weights between the hidden and output layers, as in the formula given Eq. (15).

$$o_{C-\mathrm{IFTS-FM}} = f_2\left(\left(\sum_{l=1}^{c} Hw_t^i o_i\right) + \left(\sum_{l=1}^{c}\mathrm{IW}_t\mathrm{INPUTS}(t)\right)+b_t^o\right), i=1,2,\ldots,K \tag{15}$$

f_2, in Eq. (16), is the linear activation function and $INPUTS(t)$ is a vector which is composed of inputs for the t time $(\mu_{Ll}(t-1), v_{Ll}(t-1), y_{t-j})$.

3.3 BC-IFTS-FM

In this study, until now, it has been focused on STLF, which has a univariate structure. In this univariate structure, the intuitionistic fuzzy relationships between the inputs consisting of the loads' own lagged variables, memberships, and non-memberships and the output are modelled. However, based on the fact that the load is particularly affected by the air temperature, an approach can be put forward where the air temperature information is also used in STLF. In this respect, for STLF, this study also presents a C-IFTS-FM in which air temperatures are also included in the model.

Definition 6 (*BC-IFTS-FM*) Let IF_t be an intuitionistic fuzzy time series, $I_1, I_2, \cdots, I_c$ are IFSs on the universal set, and $\mu_{I_l}(t)$, $v_{I_l}(t)$ are the degree of membership and non-membership values of the tth observation for both time series (Y_t and Z_t) in the lth IFS, respectively. The high-order BC-IFTS-FM model can be defined as below:

$$\begin{aligned} Y_t = F(&F_1(Y_{t-1}Y_{t-2}\cdots, Y_{t-m}, Z_{t-1}Z_{t-2}\cdots,\\ &Z_{t-m}, \mu_{I_1}(t-1), \mu_{I_2}(t-1), \cdots, \mu_{I_c}(t-1), v_{I_1}(t-1),\\ &v_{I_2}(t-1), \cdots, v_{I_c}(t-1)), F_2(Y_{t-1}Y_{t-2}\cdots, Y_{t-m},\\ &Z_{t-1}Z_{t-2}\cdots, Z_{t-m}, \mu_{I_1}(t-1), \mu_{I_2}(t-1), \cdots, \mu_{I_c}(t-1),\\ &v_{I_1}(t-1), v_{I_2}(t-1), \cdots, v_{I_c}(t-1))) + \varepsilon_t \end{aligned} \tag{16}$$

BC-IFTS-FM, via IFCM, generates the membership and the non-membership by clustering load and air temperatures together and also uses lagged variables of both load and air temperature as input besides these memberships and non-memberships. The basic structure of BC-IFTS-M can be given in Fig. 3.

The list of process parameters is given below:

$maxitr$ Maximum number of iterations.
m Number of lagged crisp variables of Load and Temperature.
K The number of HLU.
t_{train} Number of observations for the training set.
t_{test} Number of observations for test set.
c Number of IFSs.
T Number of time series observations.

The pseudo-code is presented which defines as a step-by-step description of the working principle of both C-IFTS-FM algorithm.

Algorithm: UC-IFTS-M / BC-IFTS-M.

Input : Observations of time series Y_t , $t = \underline{1, t_{train}}$

Output : Forecasts of time series $\hat{Y}_t$, $t = \underline{1, t_{train}}$

BEGIN

Set the parameters

Normalise Y_t / Y_t and T_t

Perform I-FCM

Generate the model's inputs

I_U: $Y_{t-j}, \mu_l(t-1), v_l(t-1)$, $j = \underline{1, m}$; $l = \underline{1, c}$ (UC-IFTS-FM)

I_B: $Y_{t-j}, T_{t-k}, \mu_l(t-1), v_l(t-1)$, $j = \underline{1, m}$; $k = \underline{1, n}$; $l = \underline{1, c}$ (BC-IFTS-FM)

$itr = 0$

Generate the parameters (weights and biases) randomly from $Uniform(-10,10)$

Repeat

$itr = itr + 1$

Calculate the outputs $\hat{Y}_t$ $t = \underline{1, t_{train}}$

Calculate the cost function $itrRMSE_r$, $r = \underline{1, pn}$

Update the parameters

Until $itr == maxitr$

Identify the best parameter values

Compute the outputs over the best parameter values $\hat{Y}_t$ $t = \underline{t_{train} + 1, T}$

END

4 Data and implementation scenarios

Short-term load data include complicated relations for forecasting. These complicated relations can be in a structure of linear or nonlinear. Moreover, most of the time, it can consist of both of these structures. In this context, the STLF problem has been chosen to validate and demonstrate the performance of the forecasting model proposed in this study on a forecasting problem with such complex relationships. Moreover, as emphasized before, for the power suppliers and managers, performing an accurate and consistent STLF is crucial. By the objectives of this study, firstly, the hourly load data of the power supply company of the city of Johor in Malaysia (PSJM) generated in 2009 and 2010 (see Fig. 4) were analysed. Furthermore, as mentioned before, based on the fact that the load is particularly affected by the air temperature, the air temperature data sets (see Fig. 5) observed in the same period were also used as auxiliary time series in STLF.

Secondly, as in Deng et.al. [46], the proposed UC-FITS-FM was applied to three data sets (HEXING, CHENGNAN, and EUNITE). All experimental data sets were downloaded at https://github.com/chenxiaoxin-102496/DataSet.git given in [46]. HEXING and CHENGNAN data sets consist of 251 (from 8/01/2019 to 04/07/2020) and 247 (from 08/05/2019 to 04/07/2020) observations, respectively. EUNITE data set consists of 761 observations (from 01/01/1997 to 01/31/1999).

Finally, the proposed univariate model's performance was evaluated using Türkiye's monthly electricity production (T-MEP) data sets. The data sets were sourced from Energy Exchange Istanbul (EXIST) (https://www.epias.com.tr/en/) and reorganized into monthly periods, comprising 125 observations from January 2012 to May 2022, just like in the study of Gulay et al. [47].

4.1 Data organization

The forecasting performance of the proposed UC-IFTS-FM and BC-IFTS-FM was validated and evaluated by dividing the data sets into two sets as training and test. For PSJM, the details of the training and test sets created for the hourly load data consisting of 8760 observations for each year are given in Fig. 6.

Moreover, for the HEXING, CHENGNAN, and EUNITE data sets, the details of the training and test sets are given in Fig. 7.

Finally, for the T-MEP data sets, the details of the training, validation, and test sets are given in Fig. 8. In these data sets, unlike the others, just like in the study of

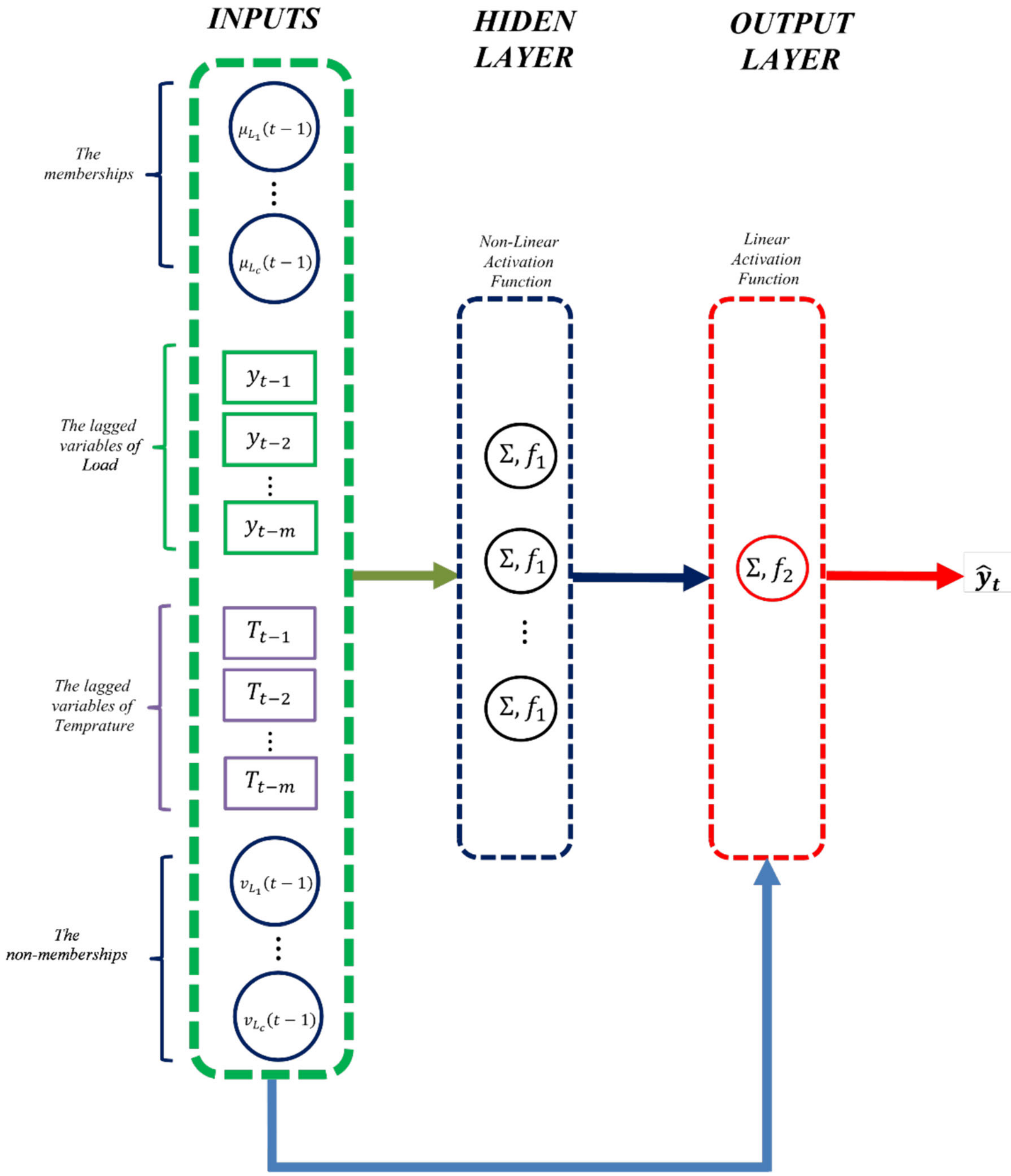


Fig. 3 The structure of BC-IFTS-M designed for STLF

Gulay et al. [47], a validation data set was used and the optimal hyperparameters were determined according to the performance of the trained models on the validation data sets. Türkiye’s electricity production consists of various components depending on the sources used for generation. The proposed univariate model was used to forecast seven components and the total production (TP) data sets. These seven data sets show how much electricity is produced from natural gas (NG), hydro (HYD), coal (COAL), wind (WIND), fuel oil (FO), bioenergy (BIO), and geothermal power (GEO).

4.2 Performance measures

The comparative performance evaluation of the models was realized by using mean absolute percentage error (MAPE) and root mean square error (RMSE) which are given in Eqs. (17) and (18).

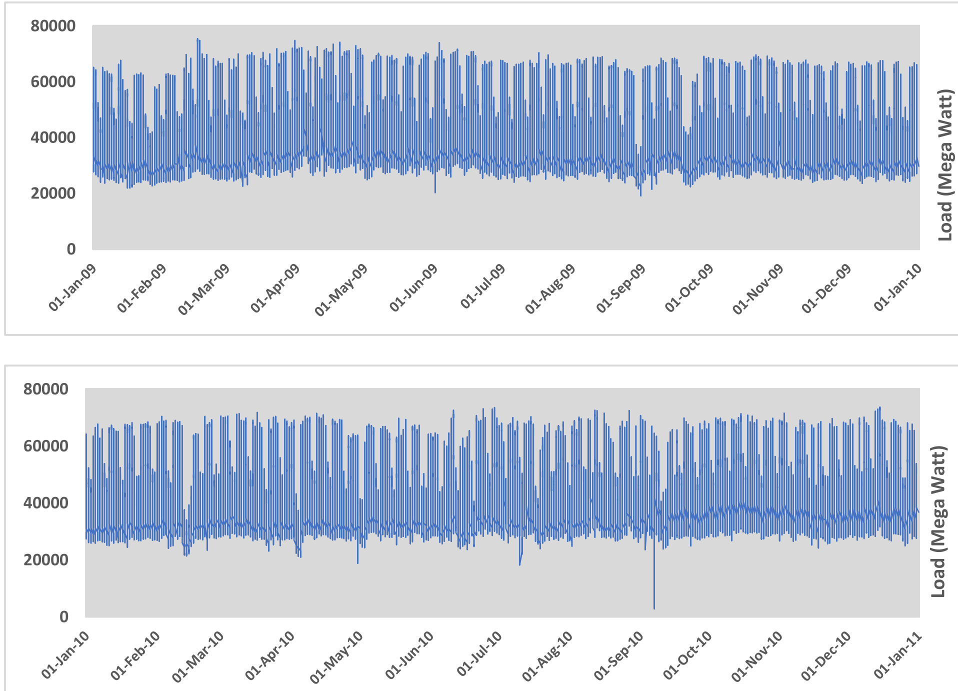


Fig. 4 The hourly load data of Malaysia of the years of 2009 and 2010

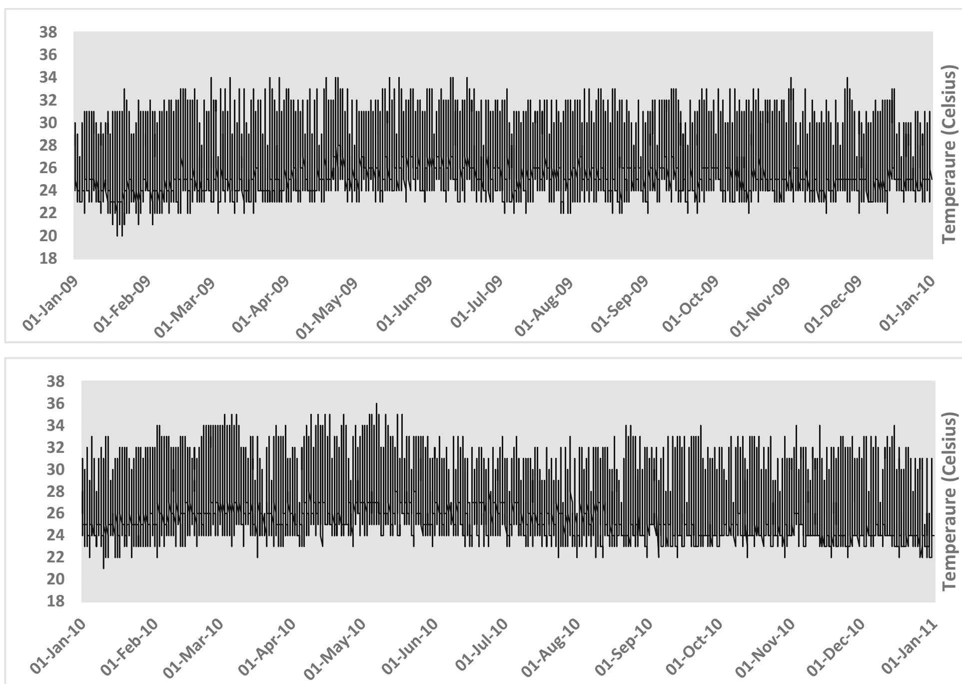


Fig. 5 The hourly air temperature data of the years of 2009 and 2010

$$\text{MAPE} = \text{mean}\left(\left|\frac{\text{Actual}_t - \text{Forecasted}_t}{\text{Actual}_t}\right| \times 100\%\right); t = 1,2,\cdots,T \quad (17)$$

$$\text{RMSE} = \sqrt{\text{mean}\left((\text{Actual}_t - \text{Forecasted}_t)^2\right)}; t = 1,2,\cdots,T \quad (18)$$

Moreover, the median relative absolute error (MdRAE), a measure of relative error, is also used to compare with a reference model. The MdRAE is calculated as in Eq. (20). Here, r_t is the relative error given in Eq. (19).

$$r_t = \frac{\text{Actual}_t - \text{Forecasted}_t}{\text{Actual}_t - \text{Forecasted}_t^*}; t = 1,2,\cdots,T \quad (19)$$

$$\text{MdRAE} = \text{Median}|r_t|; t = 1,2,\cdots,T \quad (20)$$

Forecasted^*_t is the forecasted values produced by a benchmark model chosen as the reference model. The naive model was considered as a reference model, just like most of the other studies in the literature. In the naive model; $\text{Forecasted}^*_t = \text{Actual}_{t-1}$.

4.3 Scenarios of implementations

This study basically presents two forecasting models as UC-IFTS-FM and BC-IFTS-FM. Different implementations were designed by taking different parameters for the analysis process in both forecasting models for STLF. These parameters and their usage characteristics are detailed in Table 1.

The results of 288 different applications were evaluated using statistics such as average, minimum, and maximum. In this evaluation, the worst designs in terms of performance besides the best designs were also considered and compared with the best performances of other models available in the literature. In this way, the performance of the proposed models for the forecasting problem was investigated extensively as a whole.

4.4 Consistency/reliability and validity of forecasts

As in all other forecasting problems, there are two important concepts for forecasting models in STLF: Reliability and validity. For a model, which is reliable, it is said that it has internal consistency. Actually, reliability is a principle with a degree. Therefore, one cannot mention being completely reliable of any particular model. Since chance and random effects are always present, it is quite normal for analyses to differ in performance. However, for a sufficient and efficient forecasting model, performance is expected to vary within the narrow intervals from one implementation to another.

Another important concept is validity which can be considered as a measure of accuracy for a forecasting model. An interaction between reliability and validity is explicit, and this interaction can be illustrated in Fig. 9. On the first dartboard, the hits are far from the target and quite messy. The mentioned model has a high margin of error and its consistency cannot be relied upon. The second dartboard represents an illustration of the reliable but not valid hits. Illustrated model has fairly low variability, but outputs/hits are off-target. The third one represents a case where the outputs are neither reliable nor valid. The final illustration includes the case where the expected from a satisfactory forecasting model. The outputs/hits, in this case, are both consistently showing similar values and fairly near to the targets of interest. In the light of all this information, the two proposed models were run 30 times over architectural structures that produced both the best and the worst results to be detailed examination in terms of validity and consistency.

5 Results and discussions

To demonstrate the effectiveness and forecasting ability of the proposed models, the results have been evaluated from different points of view.

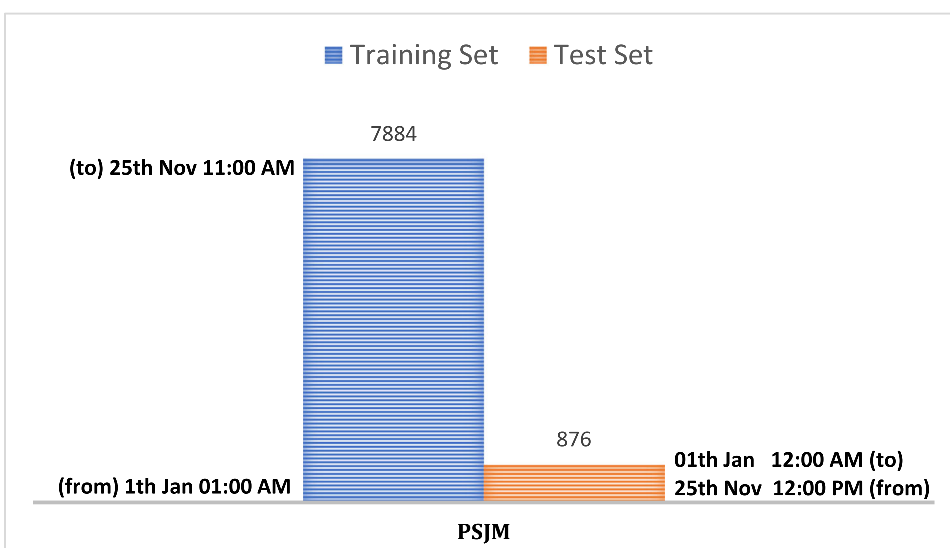


Fig. 6 Data organization for PSJM data sets in STLF

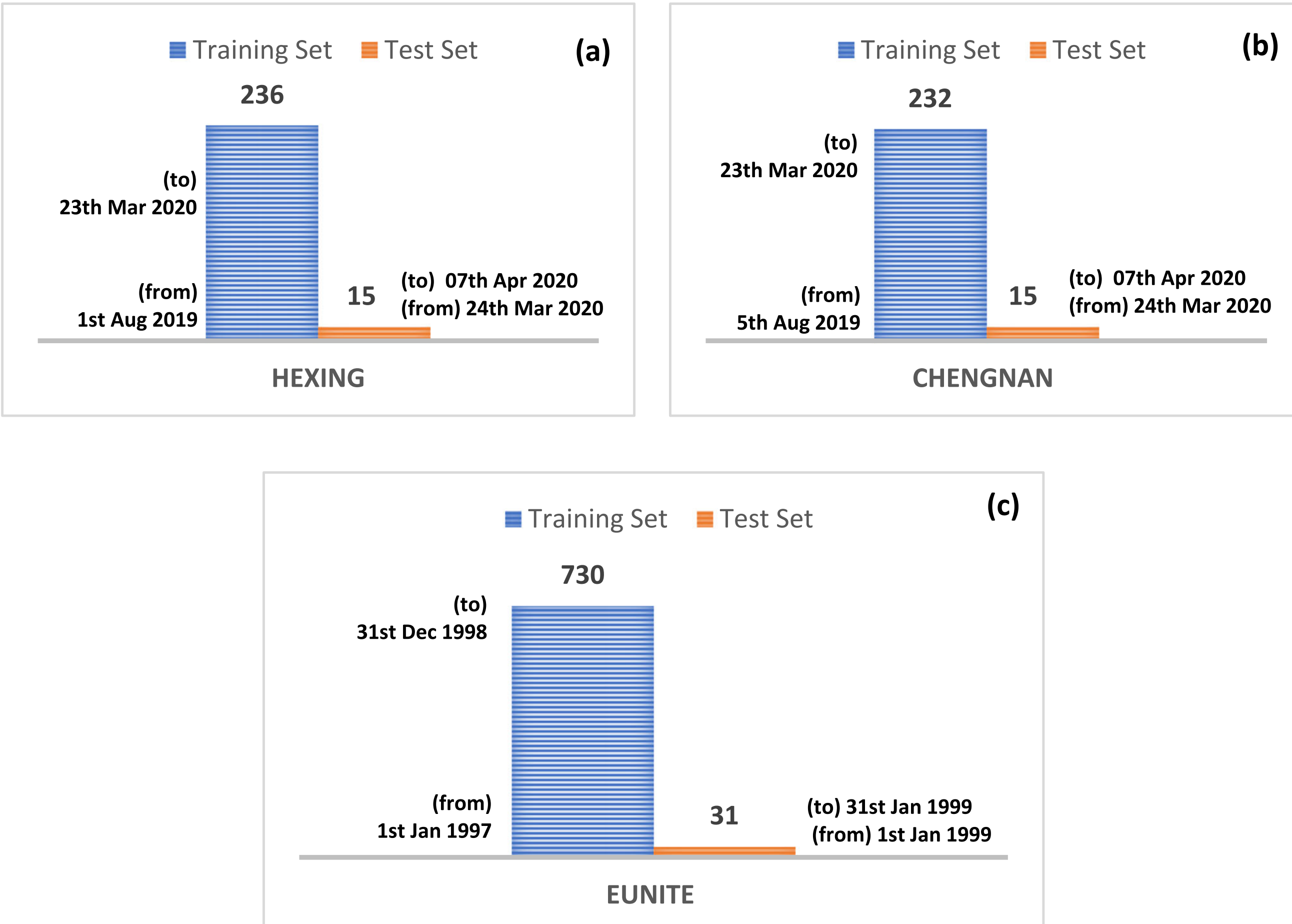
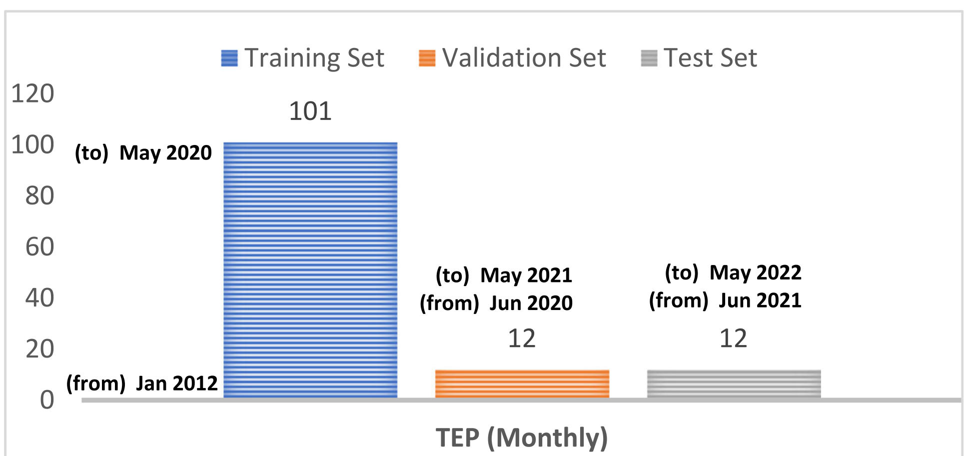


Fig. 7 Data organization for HEXING (**a**), CHENGNAN (**b**), and EUNITE (**c**) data sets

Fig. 8 Data organization for T-MEP data sets



5.1 Evaluation of all implementations-PSJM data sets

As it is emphasized before, for STLF, 288 different implementations were designed. Table 2 shows some of the statistics belonging to 3 error criteria obtained from 288 applications for the training and test sets of PSJM data sets.

From Table 2, the UC-IFTS-FM produced forecasts with a minimum MAPE value of 0.3203% for the test set of load data in the 2009 year. For the test set of load data in the 2010 year, the forecasts produced by UC-IFTS-FM had

Table 1 The implementation designs

Changing	# of Lagged Variables	# of IFS	# of HLU
from	2	3	2
to	5	10	10
# of Implementation	288		

a minimum MAPE value of 0.3571%. Another remarkable finding in Table 2 is that the model can produce forecasts with a percentage error of approximately 2% (*for 2009 load data test set: 2.6527% and for 2010 load data test set: 2.1681%*) even for the cases where it shows the worst performance in all 288 analysis. Moreover, for both the 2009 and 2010 years, load data sets, the average of the MAPE values of the forecasts obtained from 288 implementations of UC-IFTS-FM was observed to be even less than 1%. The reason why the evaluations made so far have been made over the MAPE is that MAPE is a pure error criterion independent of the scale of the data. According to Table 2, parallel comments can also be made for RMSE and MdRAE. All these findings and conclusions are also valid for BC-IFTS-FM, which produces similar results. In addition, when the results of UC-IFTS-FM and BC-IFTS-FM are evaluated together; although these two models have close forecasting performances, it can be said that BC-IFTS-FM has a slightly better performance. Another result that can be drawn from Table 2 is that the performances in the training and test sets for load data for both 2009 and 2010 are close to each other. This can be seen as an indication that the proposed models are producing consistent results, and they are also producing reliable forecasts for out-of-sample data.

5.2 Comparison to the state-of-the-art models—PSJM data sets

Among the 288 analyses, the models were run 30 times for the two cases that produced the best and the worst results in terms of RMSE values over the training set, and the average statistics of the obtained MAPE, RMSE, and MdRAE criteria values were calculated. As a traditional time series forecasting method, seasonal auto-regressive integrated moving average (SARIMA) was preferred to use in comparison because the SARIMA model has the best compatibility to the Malaysia load data sets in the years of 2009 and 2010 via their seasonal structure. The best models were determined as SARIMA (1,0,3)(2,1,2) for load data 2009 and SARIMA(1,0,1)(1,1,2) for load data in 2010 [27]. Some of the other models used in the comparative evaluation of performances are fuzzy-based models, while others are forecasting models based on deep networks. These are:

- Probabilistic weighted fuzzy time series (PWFTS)
- Weighted fuzzy time series (WFTS)
- Integrated weighted fuzzy time series (IWFTS)
- Long short-term memory (LSTM)
- Model 1 (168 former lags)
- Model 2 (72 former lags)
- Model 3 (48 former lags)
- Model 4 (24 former lags)
- The combined method (FTS-CNN)

These results are presented in Tables 3, 4 and 5 together with the results obtained with other STLF methods. The results of other STLF methods are taken from Sadaei et.al. [27]. Considering Table 3, it is observed that the proposed BC-IFTS-FM, which uses temperature data as auxiliary time series in the analysis process, showed the best performance for both the years 2009 (0.43%) and 2010 (0.41%) in the best case of it. Moreover, even in the worst case of BC-IFTS-FM, it is seen that the proposed model produced superior forecasts than the other models in the STLF literature (2.33% for 2009 and 2.06% for 2010). In addition, UC-IFTS-FM, which performs a univariate analysis process, produced better forecasts than the other current models for both the best and worst cases of it. The proposed BC-IFTS-FM, for its best cases, achieved approximately 86% better accuracy than the FTS-CNN, which ranked second in both 2009 and 2010. Compared to the FTS-CNN, which ranked second even in the worst cases of the proposed model, these rates are approximately 23% and 29% for 2009 and 2010, respectively. Similarly, the proposed UC-IFTS-FM has approximately 85% better forecasting performance than FTS-CNN, which ranked second for both 2009 and 2010, in the case of the best

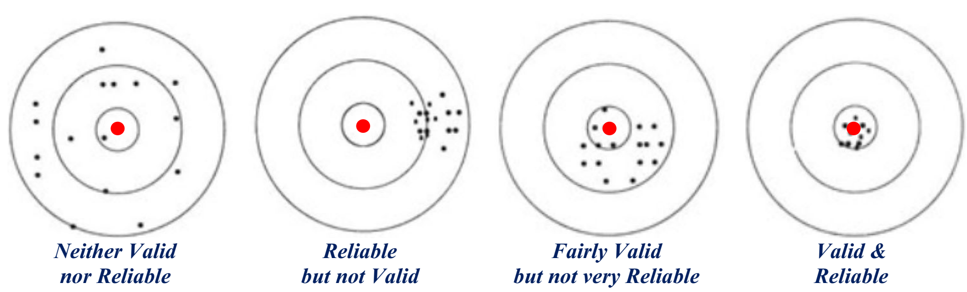


Fig. 9 The interaction between reliability/consistency and validity

Table 2 Some statistics on error criteria for 288 implementations

Model	Statistic	Data Set	MAPE		RMSE		MdRAE	
			2009	2010	2009	2010	2009	2010
UC-IFTS-FM (UNIVARIATE)	Min	Training	0.3317%	0.3766%	175.4178	211.01606	0.0463	0.0598
		Test	0.3203%	0.3571%	159.5506	210.21613	0.0459	0.0578
	Max	Training	2.5914%	2.5891%	1335.5160	1448.87486	0.4167	0.4201
		Test	2.6527%	2.1681%	1277.0889	1427.48491	0.5372	0.4380
	Average	Training	0.8819%	0.9868%	481.22116	538.73720	0.1442	0.1615
		Test	0.8842%	0.8956%	444.65748	530.84009	0.1623	0.1659
BC-IFTS-FM (BIVARIATE)	Min	Training	0.3255%	0.3806%	175.0835	205.2821	0.0472	0.0576
		Test	0.3246%	0.3444%	159.8204	210.8734	0.0522	0.0550
	Max	Training	2.6800%	2.5818%	1455.4642	1494.2805	0.4220	0.4191
		Test	2.6056%	2.3178%	1388.8959	1438.4058	0.4960	0.4014
	Average	Training	0.8555%	0.9463%	463.5468	512.2697	0.1382	0.1531
		Test	0.8673%	0.8537%	435.7309	497.9765	0.1578	0.1588

situation. Even in the worst cases of the proposed model, these rates are approximately 25% and 28% for 2009 and 2010, respectively. At this point, it should be noted that the selection of the best and the worst cases among the 288 implementations performed for the proposed models is determined based on the performance of the models over the training set. That is, the best case implementations determined here can always be obtained without the need for out-of-sample data points. The worst results are also presented to emphasize the superior performance of the proposed methods in each case.

Similar findings to those obtained for MAPE were also obtained for RMSE. Both of the proposed models, for their best cases, presented over 85% better accuracy than the FTS-CNN, which ranked second in both 2009 and 2010. In addition, in terms of the RMSE criterion, the proposed models, even for their worst cases, performed over 30% and 20% better accuracy than the FTS-CNN in 2009 and 2010, respectively.

Considering the MdRAE criterion, both of the proposed models, for their best cases, presented over 80% better performance than the FTS-CNN, which ranked second in both 2009 and 2010. Moreover, for their worst cases compared to FTS-CNN in 2009, the proposed BC-IFTS-FM made progress with a ratio of 5%, while the proposed UC-IFTS-FM made progress with a ratio of 10%. For 2010, these progress levels are about 15% for both proposed models.

5.3 The statistical evaluation of the errors—PSJM data sets

By running the proposed models for STLF 30 times, it can make a statistical evaluation over some statistics of the errors. Table 6 presents these statistics for MdRAE error criterion values produced by BC-IFTS-FM and UC-IFTS-FM for test sets, their best cases. It is clear from Table 6 that both proposed models produce forecasts with quietly low MdRAE values compared to other models, even for repetition where forecasts are generated with a maximum MdRAE. In addition, when the standard deviation is taken into account, it is observed that the variation of the forecasting errors is pretty small and the very narrow confidence interval (CI) bounds also support this situation. In addition, considering the kurtosis and skewness values of the errors and the standard errors of these values, it was determined that the MdRAE values showed a normal distribution and thus the error probabilities could be calculated.

The distribution of MdRAE values obtained from 30 repetitions of both models can be presented with the graphs given in Figs. 10 and 11. The MdRAE values which have such scattering can be evaluated in two different ways. One of them is reliability. For a reliable forecasting model, performance metrics are expected to vary within narrow ranges from one implementation to the other, as mentioned in previous sections. From Fig. 10, in the case of the best situations, it is observed that the MdRAE values of forecasts produced by two proposed models are scattered between 0.04 and 0.10 for the years 2009 and 2010. From Fig. 11, in the case of the worst situations, it is seen that the MdRAE values are scattered approximately between 0.20

Table 3 Average MAPE (%) of the models in STLF

Forecasting model	2009		2010	
	Average	Rank	Average	Rank
BC-IFTS-FM (The Proposed Model/Best)	**<u>0.43</u>**	**<u>1</u>**	**<u>0.41</u>**	**<u>1</u>**
BC-IFTS-FM (The Proposed Model/Worst)	2.33	4	2.06	3
UC-IFTS-FM (The Proposed Model/Best)	**0.45**	**2**	**0.44**	**2**
UC-IFTS-FM (The Proposed Model/Worst)	2.25	3	2.07	4
SARIMA	4.68	12	4.23	11
FTS-CNN	3.02	5	2.89	5
PWFTS model 1	3.86	7	4.00	8
PWFTS model 2	4.59	11	5.83	12
WFTS	5.33	14	9.09	14
IWFTS	5.32	13	7.69	13
LSTM model 1	3.71	6	3.45	6
LSTM model 2	4.55	10	4.21	9
LSTM model 3	4.11	9	4.23	10
LSTM model 4	3.93	8	3.88	7

Bold-Underline values represent the best-performing models

Bold values only represent the second-best-performing models

Table 4 Average RMSE of the models in STLF

Forecasting model	2009		2010	
	Average	Rank	Average	Rank
BC-IFTS-FM (The Proposed Model/Best)	**<u>200.41</u>**	**<u>1</u>**	**<u>228.28</u>**	**<u>1</u>**
BC-IFTS-FM (The Proposed Model/Worst)	1214.03	4	1296.37	3
UC-IFTS-FM (The Proposed Model/Best)	**213.98**	**2**	**251.30**	**2**
UC-IFTS-FM (The Proposed Model/Worst)	1184.08	3	1340.95	4
SARIMA	2763.66	11	2501.25	11
FTS-CNN	1777.99	5	1702.70	5
PWFTS model 1	2230.91	7	2162.57	8
PWFTS model 2	2987.40	14	3797.35	12
WFTS	2930.36	12	4419.11	13
IWFTS	2961.17	13	4663.17	14
LSTM model 1	2194.19	6	2037.49	6
LSTM model 2	2689.42	10	2044.68	7
LSTM model 3	2413.60	9	2483.71	10
LSTM model 4	2317.88	8	2279.23	9

Bold-Underline values represent the best-performing models

Bold values only represent the second-best-performing models

and 0.60 for both years. In both cases, it has been observed that the MdRAE values are scattered within a fairly narrow range, which is proof of the reliability of the proposed models. Secondly, to focus on validity. Even in the worst case, the proposed models revealed a very small MdRAE in all repetitions. Among repetitions, the highest MdRAE values have been found around 0.10 for the best cases and 0.60 for the worst case. These values are evidence that the proposed models produce highly satisfactory predictive results in all cases and are valid models.

5.4 The convergence time of the proposed models—PSJM data sets

This subsection is aimed to prove that the proposed models have a reasonable convergence time as another finding proving their superiority over other existing models. For this purpose, the average convergence times for 30 times repeated implementations in the best and worst cases of the two models were determined in seconds. The information collected in this direction is summarized in Table 7. From Table 7, it is seen that the convergence time is longer than

Table 5 Average MdRAE of the models in STLF

Forecasting model	2009		2010	
	Average	Rank	Average	Rank
BC-IFTS-FM (The Proposed Model/Best)	**<u>0.0707</u>**	**<u>1</u>**	**<u>0.0712</u>**	**<u>1</u>**
BC-IFTS-FM (The Proposed Model/Worst)	0.4621	4	0.3869	4
UC-IFTS-FM (The Proposed Model/Best)	**0.0720**	**2**	**0.0756**	**2**
UC-IFTS-FM (The Proposed Model/Worst)	0.4358	3	0.3853	3
SARIMA	0.7504	12	0.6655	10
FTS-CNN	0.4857	5	0.4537	5
PWFTS model 1	0.6107	8	0.6319	8
PWFTS model 2	0.5284	6	0.7638	12
WFTS	1.0838	14	2.1105	14
IWFTS	1.0780	13	1.2185	13
LSTM model 1	0.5981	7	0.5435	6
LSTM model 2	0.7313	11	0.6622	9
LSTM model 3	0.6610	10	0.6673	11
LSTM model 4	0.6273	9	0.5991	7

Bold-Underline values represent the best-performing models

Bold values only represent the second-best-performing models

Table 6 Some statistics on MdRAE error criterion of test sets for 30 times running

Model	Statistic		MdRAE	
			2009	2010
C-IFTS-FM (UNIVARIATE)		Min	0.0459	0.0500
		Max	0.0912	0.1006
		Standard Deviation	0.0146	0.0127
	Mean	Lower Bound of %95 CI	0.0665	0.0709
		Value	0.0720	0.0756
		Standard Error	0.0027	0.0023
		Upper Bound of %95 CI	0.0774	0.0804
		Skewness	− 0.3442	0.2143
		Standard Error	0.4268	0.4270
		Kurtosis	− 1.3805	− 0.4560
		Standard Error	0.8326	0.8330
BC-IFTS-FM (BIVARIATE)		Min	0.0471	0.0561
		Max	0.0952	0.0858
		Standard Deviation	0.0144	0.0079
	Mean	Lower Bound of %95 CI	0.0653	0.0682
		Value	0.0707	0.0712
		Standard Error	0.0026	0.0014
		Upper Bound of %95 CI	0.0761	0.0741
		Skewness	0.1018	0.1379
		Standard Error	0.4269	0.4270
		Kurtosis	− 1.0479	− 0.7904
		Std. Error	0.8327	0.8330

the others in cases the best results are produced. This is because the architectures which produced the best performances have more IFS and HLUs. Even for these cases, the convergence time is on average between 1.5 and 2.5 min, and such a convergence time is remarkably reasonable and applicable. In the worst scenarios, the convergence speed is much higher, with around 5 s or less. Given forecasts having high accuracy obtained for these scenarios, it can be

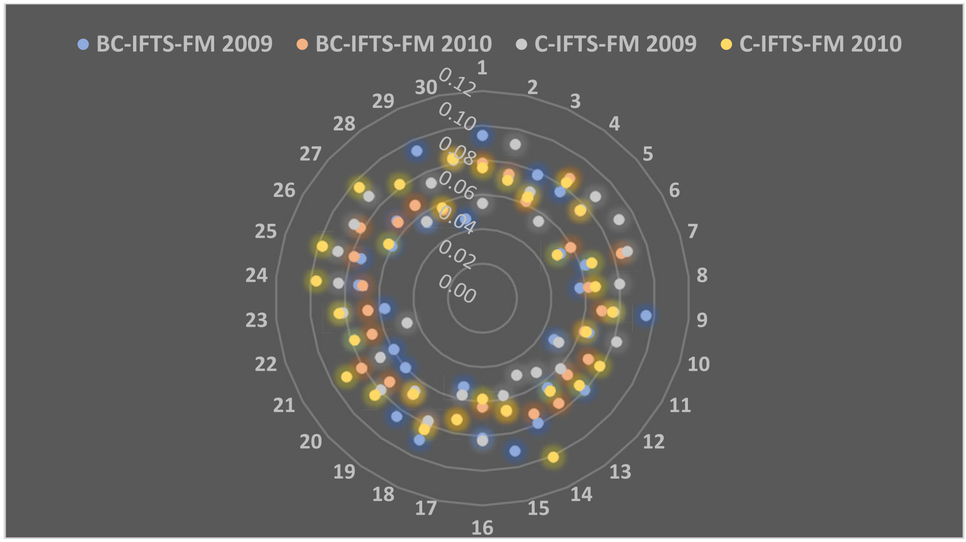


Fig. 10 The distribution of MdRAE values in the case of the best situations

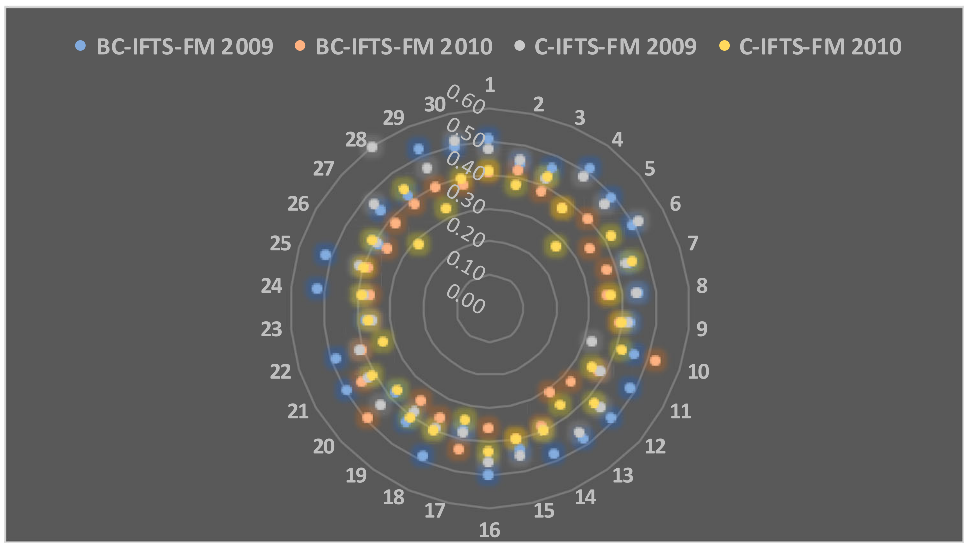


Fig. 11 The distribution of MdRAE values in the case of the worst situations

said that these times are outstanding for STLF practitioners.

5.5 The fit of forecasted and observed data points—PSJM data sets

It has been proven from different points of view that both models proposed for both data sets in all scenarios have superior performance compared to other models until this section. In this section, these findings are supported by providing visual evidence showing the high fitting between the forecasts obtained for some scenarios and the observed values. For this purpose, in the best situations of BC-IFTS-FM, the graph of the forecasts and observations values is shown in Figs. 12 and 13. The high synchronization of the forecasts with the observed values is clearly observed in these two graphs.

5.6 Comparison to the state-of-the-art models—HEXING, CHENGNAN, and EUNITE data sets

For these data sets, the best results, in terms of RMSE values over the training set, were compared with some other methods used by Deng et.al. [46]. Other methods can be listed as:

The comparative evaluation of performances are fuzzy-based models, while others are forecasting models based on deep networks. These are:

- Gene expression programming (GEP)
- Gene expression programming based on least square (GEP-LS)
- Gene expression programming based on K-means and least square (GEP-KLS)
- Quantitative combination load forecasting (QCLF)

Table 7 Convergence time (in seconds) of the proposed models

The Proposed Model	Year of Data	Case	# of Lagged Variables	# of IFS	# of HLU	Time in averagely second
BC-IFTS-FM	2009	Best	3	9	8	120.55
		Worst	4	3	2	5.63
	2010	Best	2	10	7	146.40
		Worst	3	3	2	5.41
UC-IFTS-FM	2009	Best	2	10	10	88.91
		Worst	3	3	2	3.91
	2010	Best	2	10	9	94.28
		Worst	4	3	2	3.83

- LSTM
- Pattern-based local linear regression (PLLR)
- Back-propagation neural network (BPNN)

All results of other methods are from Deng et al. [46]. The comparative results, in terms of RMSE and MAPE metrics, are presented in Tables 8 and 9.

Table 8 shows that the proposed UC-IFTS-FM produced the best performance for all three data sets, even for the worst case of the proposed model over other methods. For the best case, the forecasts produced by the proposed UC-IFTS-FM had RMSE values of 0.5747, 0.2907, and 1.4992, respectively. These values were 8.12, 4.80, and 0.84, even in the worst case. Even in the worst case, the proposed model made progress by about 55% (*HEXING*), 70% (*CHENGNAN*), and 39% (*EUNITE*), respectively, over the best of the others. Also, in the best case, progress was phenomenal, with 95%, 97%, and 86%, respectively, over the best of the rest. To respecify, the best and the worst cases for testing sets among all implementations performed for the proposed model were determined based on the performance of the models over the training set.

Similar results, for the MAPE, to the RMSE metric were obtained. In HEXING forecasting, the proposed model, in its best case, presented 93% better accuracy than BPNN which was the best among the others. In CHENGNAN and UENITE forecasting, the accuracy progress levels are 55% and 37% over the GEP-KLS which was the best among the others, respectively.

5.7 Comparison to the state-of-the-art models–T-MEP data sets

The optimal hyperparameter combinations obtained from the validation sets for the eight-time series that make up the T-MEP data set are given in Table 10.

Tables 11 and 12 offer the MAPE and RMSE, along with their ranks (R), as measures for assessing and comparing the forecasting performance of the models on out-of-sampling time points. The loss function values of other models used in the comparison were sourced from the study by Gulay et al. [47]. The results produced by 19 different forecasting models in addition to the proposed univariate model were presented to compare prediction performances in Tables 16 and 17. However, since the ARDL model has its own characteristics, the values corresponding to certain series are left blank, as in the study of Gulay et al. [47]. The MAPE results presented in Table 11

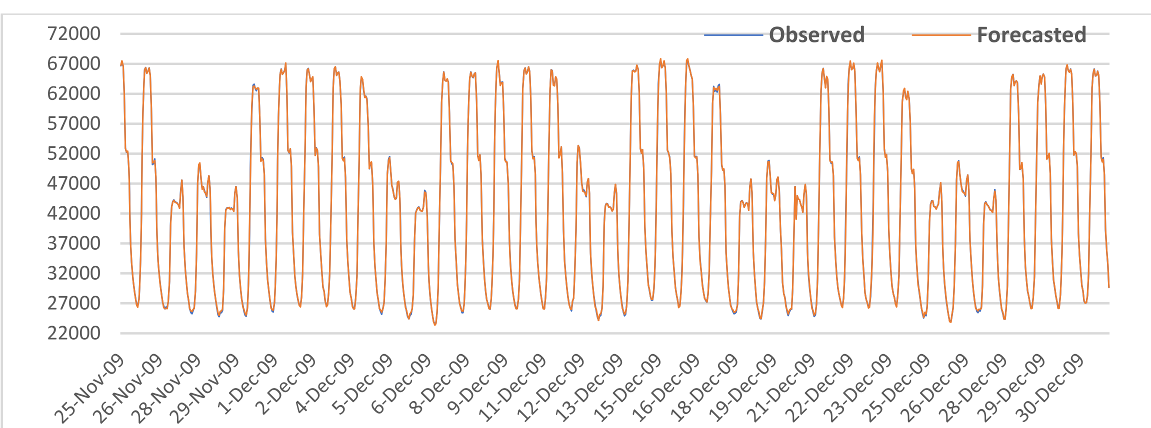


Fig. 12 The harmony of forecasted and observed loads for BC-IFTS-FM in 2009

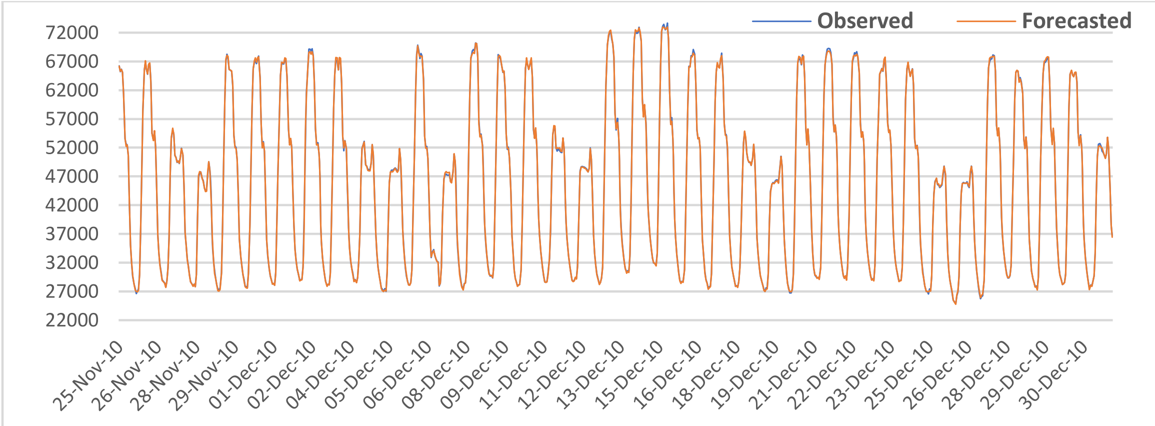


Fig. 13 The harmony of forecasted and observed loads for BC-IFTS-FM in 2010

reveal that the proposed UC-IFTS-FM model, compared to the others, demonstrates outstanding forecasting performance for all time series. In addition, the improvements made by the proposed model in prediction performance compared to the best model among the other 19 models are also given in Table 11 for each data set. According to these results, in terms of the MAPE criterion, the proposed model has improved forecasting performance for the "COAL" data set by over 80%. The progress levels, exhibited by the proposed UC-IFTS-FM, for "HYD" and "WIND" data sets, were observed as 63.53 and 54.48%, respectively. Moreover, the progress levels, shown by the proposed UC-IFTS-FM, for "NG" and "GEO" data sets, were seen as over 40 and 30%, respectively. Furthermore, for the "TP", the progress level was over 20%. The RMSE results summarized in Table 12 prove that the proposed UC-IFTS-FM model, compared to the others, exhibits again superior forecasting performance for all time series except the "NG" data set. In terms of RMSE, the progress levels, provided by the proposed UC-IFTS-FM, for "HYD" and "COAL" data sets, reached 70 and 90% levels, respectively. Moreover, the compatibility of the forecasts obtained for each T-MEP data set with the actual observation values is visually presented in Fig. 14.

6 Conclusions

Two new forecasting models, UC-IFTS-FM and BC-IFTS-FM, were proposed in this study with the aim of short-term load forecasting. The proposed models are based on IFSs and neural networks with the cascade structure. One of these proposed models has been designed in a way that air temperature information is also used in the STLF, based on the fact that the load is particularly affected by the air temperature. The proposed models have been performed in different implementation scenarios. PSJM data sets were forecasted in 288 different scenarios with both proposed

Table 8 Performance comparison between different models for testing data sets—RMSE

Forecasting Model	Data sets					
	HEXING		CHENGNAN		EUNITE	
	RMSE	Rank	RMSE	Rank	RMSE	Rank
The Proposed UC-IFTS-FM (Best)	**0.5747**	**1**	**0.2907**	**1**	**1.4992**	**1**
The Proposed UC-IFTS-FM (Worst)	**5.6477**	**2**	**3.4904**	**2**	**6.7035**	**2**
QCLF	12.9326	5	12.5342	4	13.9355	6
GEP-KLS	12.7206	4	11.5153	3	10.9587	3
GEP-LS	21.5100	9	22.5342	9	14.4026	8
GEP	12.9326	6	12.5342	5	13.9355	7
LSTM	18.7941	8	14.7829	8	11.9168	4
PLLR	15.7051	7	12.9215	6	15.8813	9
BPNN	12.5609	3	13.6523	7	13.7958	5

Bold-Underline values represent the best-performing models

Bold values only represent the second-best-performing models

Table 9 Performance comparison between different models for testing data sets—MAPE(%)

Forecasting Model	Data sets					
	HEXING		CHENGNAN		EUNITE	
	MAPE	Rank	MAPE	Rank	MAPE	Rank
The Proposed UC-IFTS-FM (Best)	**0.90**	**1**	**0.33**	**1**	**0.18**	**1**
The Proposed UC-IFTS-FM (Worst)	**8.12**	**2**	**4.80**	**2**	**0.84**	**2**
QCLF	13.90	4	13.55	5	1.83	6
GEP-KLS	13.99	6	11.23	3	1.33	3
GEP-LS	20.37	9	23.55	9	1.83	8
GEP	13.90	5	13.55	6	1.83	7
LSTM	18.38	8	12.55	4	1.56	4
PLLR	15.08	7	14.29	8	2.21	9
BPNN	12.93	3	14.17	7	1.72	5

Bold-Underline values represent the best-performing models

Bold values only represent the second-best-performing models

Table 10 Optimal hyperparameter values for T-MEP data sets

Hyperparameter	Data Sets							
	TP	NG	HYD	COAL	WIND	FO	GEO	BIO
# of Lagged Variables	2	2	2	2	2	2	3	2
# of IFS	8	10	10	10	4	6	5	3
# of HLU	3	2	7	5	6	5	8	7

Table 11 Performance comparison for T-MEP testing data sets—MAPE(%)

Methods	TP	*R*	NG	*R*	HYD	*R*	COAL	*R*	WIND	*R*	FO	*R*	GEO	*R*	BIO	*R*
SARIMA	2.26	6	10.68	6	10.40	4	11.13	11	14.01	11	161.23	8	2.18	3	2.80	7
HW Add	30.11	19	32.25	15	26.57	13	27.12	17	70.76	18	4573.98	17	88.67	19	88.69	19
HW Mult	30.11	20	32.27	16	26.21	12	25.97	16	65.56	17	4856.85	18	88.62	18	88.68	17
ETS	1.93	4	11.13	7	12.22	7	9.40	7	12.70	9	240.21	11	2.42	5	2.99	8
TBATS	2.27	7	12.87	8	11.82	6	10.14	8	12.64	8	124.30	7	3.67	9	3.34	12
STL-SARIMA	2.06	5	8.07	3	27.43	14	7.27	3	11.56	4	358.14	13	2.67	6	2.59	6
STL-HW Add	29.99	17	34.85	17	28.57	17	41.80	20	77.17	20	4001.27	16	87.69	16	88.58	16
STL-HW Mult	30.02	18	35.74	18	32.02	19	36.08	19	74.89	19	6990.74	19	88.14	17	88.69	18
STL-ETS	1.63	3	8.65	5	11.01	5	8.10	5	11.80	5	557.87	15	4.11	10	2.07	3
ANN	5.57	13	27.84	13	24.92	11	14.97	15	16.87	14	230.73	10	54.99	15	2.54	5
LSTM	3.91	9	17.16	10	21.20	10	10.29	9	15.81	12	76.33	6	3.07	7	3.17	11
CNN	6.93	16	30.16	14	27.56	15	13.15	13	16.20	13	166.88	9	3.60	8	2.32	4
STL-ANN	5.75	14	38.92	19	64.85	20	28.08	18	30.79	16	9909.66	20	46.85	14	3.10	10
STL-LSTM	1.63	2	4.71	2	9.13	2	7.51	4	10.42	3	50.61	5	2.09	2	3.05	9
STL-CNN	5.51	12	15.02	9	10.37	3	5.65	2	10.26	2	337.96	12	2.39	4	1.97	2
MV-ANN	5.24	10	8.14	4	12.62	8	10.61	10	13.69	10	25.63	2	9.78	13	8.76	14
MV-LSTM	5.43	11	22.65	11	29.14	18	9.12	6	12.26	7	26.60	3	5.55	11	4.41	13
MV-CNN	6.74	15	26.41	12	27.70	16	13.42	14	17.60	15	27.89	4	5.85	12	11.76	15
ARDL	3.09	8	–	–	16.54	9	12.91	12	11.90	6	403.62	14	–		–	–
UC-IFTS-FM	**1.30**	**1**	**2.65**	**1**	**3.33**	**1**	**0.78**	**11**	**4.67**	**1**	**23.58**	**1**	**1.37**	**1**	**1.50**	**1**
Progress Level	**20.25%**		**43.74%**		**63.53%**		**86.19%**		**54.48%**		**8.00%**		**34.45%**		**23.86%**	

Bold-Italic values represent the progress levels

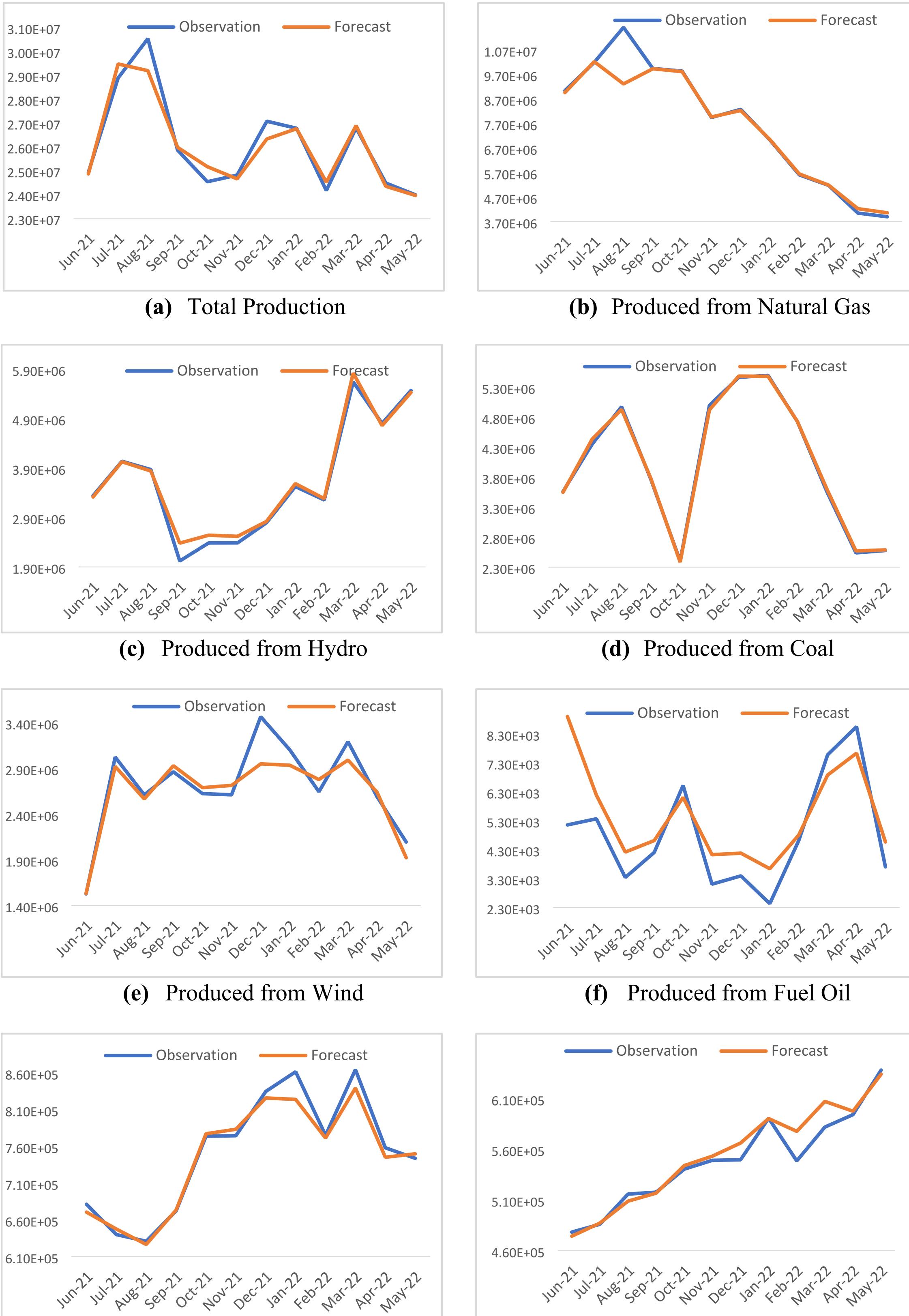
Bold-Underline values represent the best-performing models

Table 12 Performance comparison for T-MEP testing data sets—RMSE

Methods	TP	*R*	NG	*R*	HYD	*R*	COAL	*R*	WIND	*R*	FO	*R*	GEO	*R*	BIO	*R*
SARIMA	913,751	7	1,139,236	7	521,572	3	1,029,488	11	493,040	13	21,366	13	22,115	5	18,577	6
HW Add	8,208,756	19	2,375,007	13	1,881,202	16	2,578,660	18	2,000,264	18	204,925	17	669,103	19	488,318	18
HW Mult	8,209,099	20	2,378,611	14	1,885,108	17	2,492,407	17	1,883,016	17	226,518	18	668,737	18	488,314	17
ETS	685,158	4	1,076,002	6	731,128	7	823,965	6	396,434	8	28,695	14	21,670	2	19,234	8
TBATS	764,334	6	1,161,169	8	691,392	6	995,496	10	386,006	6	13,971	9	30,403	8	22,752	12
STL-SARIMA	745,460	5	867,286	4	1,485,190	10	736,196	4	333,149	4	16,426	10	22,959	6	15,310	4
STL-HW Add	8,183,556	17	2,514,487	17	2,143,232	18	3,568,360	19	2,150,919	20	191,952	16	661,969	16	487,804	16
STL-HW Mult	8,191,133	18	2,586,084	18	2,570,716	19	3,618,517	20	2,085,589	19	363,411	19	665,200	17	488,424	19
STL-ETS	675,535	3	882,909	5	638,401	4	750,137	5	341,218	5	29,168	15	34,102	9	13,104	2
ANN	1,927,761	13	2,079,753	11	1,683,690	14	1,233,064	15	509,480	14	9656	7	414,717	14	19,865	9
LSTM	1,510,685	9	1,864,405	10	1,500,310	12	875,692	8	456,276	11	4233	6	29,172	7	19,896	10
CNN	2,466,245	15	2,418,549	15	1,672,880	13	1,106,566	12	475,691	12	10,820	8	35,437	10	16,427	5
STL-ANN	1,900,929	12	3,330,084	19	2,984,626	20	2,466,662	16	873,603	16	503,645	20	434,638	15	21,385	11
STL-LSTM	658,631	2	**529,115**	**1**	450,056	2	692,133	3	328,497	3	2819	5	22,040	4	18,747	7
STL-CNN	1,707,045	10	1,276,826	9	668,990	5	587,653	2	304,432	2	20,128	11	21,767	3	13,591	3
MV-ANN	1,777,954	11	800,185	3	752,801	8	957,319	9	421,917	9	1737	3	90,018	13	57,790	14
MV-LSTM	2,052,424	14	2,236,366	12	1,485,294	11	867,887	7	422,323	10	1736	2	47,729	11	35,696	13
MV-CNN	2,575,040	16	2,448,352	16	1,710,791	15	1,176,451	14	520,575	15	1810	4	56,465	12	68,785	15
ARDL	966,049	8	–	–	934,517	9	1,161,092	13	390,786	7	20,421	12	–	–	–	–
UC-IFTS-FM	**524,296**	**1**	669,827	2	**133,672**	**1**	**39,526**	**1**	**186,479**	**1**	**1323**	**1**	**14,763**	**1**	**12,560**	**1**
Progress Level	**20.40%**		**--**		**70.30%**		**93.27%**		**38.75%**		**23.79%**		**31.87%**		**4.15%**	

Bold-Italic values represent the progress levels

Bold-Underline values represent the best-performing models



(a) Total Production

(b) Produced from Natural Gas

(c) Produced from Hydro

(d) Produced from Coal

(e) Produced from Wind

(f) Produced from Fuel Oil

(g) Produced from Geothermal

(h) Produced from Bioenergy

Fig. 14 The harmony of forecasted and observed values for UC-IFTS-FM univariate and multivariate models, and superior performance was observed compared to other models even in scenarios where the worst results were produced. Moreover, the consistency and reliability of the forecasts were proven by running the proposed models 30 times, and the results were analysed statistically. In addition, HEXING, CHENGNAN, and EUNITE data sets were analysed in 288 different scenarios with the proposed univariate model, and it was observed that even in the worst-performing scenario, the proposed model exhibited a significantly superior prediction performance compared to the current value models. Moreover, the T-MEP data sets, consisting of 8 different time series, were analysed with the proposed univariate model. In T-MEP analysis, hyperparameter tuning was performed and the best hyperparameter combination was determined based on validation sets. The main conclusions obtained at the end of the study can be summarized as follows:

- The proposed UC-IFTS-FM and BC-IFTS-FM produced superior and more accurate forecasts than current models in STLF.
- With close and successful performance demonstrated for training and test data in different scenarios, it has been proven that the proposed models do not contain the problem of overfitting.
- The proposed BC-IFTS-FM results proved the importance of info on temperature data to improve performance in STLF.
- IFS-based models also take into account the nonmemberships based on hesitation degrees. By this means, IFS-based models produced more advantageous results in STLF compared to fuzzy-based models.
- The proposed models have improved the performance in the STLF since they can simultaneously model the linear and nonlinear relationships between inputs and outputs, thanks to the cascade structure.
- The proposed models do not experience any loss of information as they do not need the time series to be converted into an image and thus exhibit superior performance.
- The proposed models achieve superior forecasting performance in all scenarios at extremely reasonable convergence times.

All findings of this paper can serve as a foundation for further studies. In particular, new approaches incorporating fuzzy sets with different properties and neural networks with various structures can build on these results. We believe these findings can be applied to mid-term and long-term time series forecasting problems in many scientific and practical areas, in addition to enhancing short-term load forecasting.

Author contributions Ozge Cagcag Yolcu contributed to conceptualization, methodology, software, and writing—original draft preparation. Hak-Keung Lam contributed to software, data curation, writing—reviewing and editing, and supervision. Ufuk Yolcu contributed to visualization, investigation, software, and validation.

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Data availability The data sets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

References

1. Yolcu OC, Egrioglu E, Bas E, Yolcu U (2022) Multivariate intuitionistic fuzzy inference system for stock market prediction: the cases of Istanbul and Taiwan. Appl Soft Comput. https://doi.org/10.1016/j.asoc.2021.108363
2. Vähäkyla P, Hakonen E, Léman P (1980) Short-term forecasting of grid load using Box-Jenkins techniques. Int J Electr Power Energy Syst. https://doi.org/10.1016/0142-0615(80)90004-6
3. Moghaddas-Tafreshi SM, Farhadi M (2008) A linear regression-based study for temperature sensi.wity analysis of iran electrical load. In: Proceedings of the IEEE international conference on industrial technology
4. Zheng Z, Chen H, Luo X (2019) A Kalman filter-based bottom-up approach for household short-term load forecast. Appl Energy. https://doi.org/10.1016/j.apenergy.2019.05.102
5. Hippert HS, Pedreira CE, Souza RC (2000) Combining neural networks and ARIMA models for hourly temperature forecast. In: Proceedings of the international joint conference on neural networks
6. Weron R (2006) Modeling and forecasting electricity loads and prices: a statistical approach. Wiley, London. https://doi.org/10.1002/9781118673362
7. Taylor JW, McSharry PE (2007) Short-term load forecasting methods: an evaluation based on European data. IEEE Trans Power Syst. https://doi.org/10.1109/TPWRS.2007.907583

8. Zhang R, Dong ZY, Xu Y et al (2013) Short-term load forecasting of Australian national electricity market by an ensemble model of extreme learning machine. IET Gener Transm Distrib. https://doi.org/10.1049/iet-gtd.2012.0541
9. Senjyu T, Takara H, Uezato K, Funabashi T (2002) One-hour-ahead load forecasting using neural network. IEEE Trans Power Syst. https://doi.org/10.1109/59.982201
10. Nowotarski J, Liu B, Weron R, Hong T (2016) Improving short term load forecast accuracy via combining sister forecasts. Energy. https://doi.org/10.1016/j.energy.2015.12.142
11. Li S, Wang P, Goel L (2016) A novel wavelet-based ensemble method for short-term load forecasting with hybrid neural networks and feature selection. IEEE Trans Power Syst. https://doi.org/10.1109/TPWRS.2015.2438322
12. Chaturvedi DK, Sinha AP, Malik OP (2015) Short term load forecast using fuzzy logic and wavelet transform integrated generalized neural network. Int J Electr Power Energy Syst. https://doi.org/10.1016/j.ijepes.2014.11.027
13. Zahedi G, Azizi S, Bahadori A et al (2013) Electricity demand estimation using an adaptive neuro-fuzzy network: a case study from the Ontario province - Canada. Energy. https://doi.org/10.1016/j.energy.2012.10.019
14. Ghofrani M, Ghayekhloo M, Arabali A, Ghayekhloo A (2015) A hybrid short-term load forecasting with a new input selection framework. Energy. https://doi.org/10.1016/j.energy.2015.01.028
15. Khan IU, Javaid N, Taylor CJ, et al (2020) Big data analytics based short term load forecasting model for residential buildings in smart grids. In: IEEE INFOCOM 2020 - IEEE conference on computer communications workshops, INFOCOM WKSHPS 2020
16. Atef S, Eltawil AB (2020) Assessment of stacked unidirectional and bidirectional long short-term memory networks for electricity load forecasting. Electric Power Syst Res. https://doi.org/10.1016/j.epsr.2020.106489
17. Kumar S, Hussain L, Banarjee S, Reza M (2018) Energy load forecasting using deep learning approach-LSTM and GRU in spark cluster. In: Proceedings of 5th international conference on emerging applications of information technology, EAIT 2018
18. Yu HK (2005) Weighted fuzzy time series models for TAIEX forecasting. Physica A Stat Mech Appl. https://doi.org/10.1016/j.physa.2004.11.006
19. Cagcag Yolcu O, Alpaslan F (2018) Prediction of TAIEX based on hybrid fuzzy time series model with single optimization process. Appl Soft Comput J. https://doi.org/10.1016/j.asoc.2018.02.007
20. Trierweiler Ribeiro G, Alves Portela Santos A, Cocco Mariani V, dos Santos CL (2021) Novel hybrid model based on echo state neural network applied to the prediction of stock price return volatility. Expert Syst Appl. https://doi.org/10.1016/j.eswa.2021.115490
21. Ren S, Wang X, Zhou X, Zhou Y (2023) A novel hybrid model for stock price forecasting integrating Encoder Forest and Informer. Expert Syst Appl. https://doi.org/10.1016/j.eswa.2023.121080
22. Sadaei HJ, Enayatifar R, Abdullah AH, Gani A (2014) Short-term load forecasting using a hybrid model with a refined exponentially weighted fuzzy time series and an improved harmony search. Int J Electr Power Energy Syst. https://doi.org/10.1016/j.ijepes.2014.04.026
23. Sadaei HJ, Guimarães FG, José da Silva C et al (2017) Short-term load forecasting method based on fuzzy time series, seasonality and long memory process. Int J Approx Reason. https://doi.org/10.1016/j.ijar.2017.01.006
24. Efendi R, Ismail Z, Deris MM (2015) A new linguistic out-sample approach of fuzzy time series for daily forecasting of Malaysian electricity load demand. Appl Soft Comput J. https://doi.org/10.1016/j.asoc.2014.11.043
25. Enayatifar R, Sadaei HJ, Abdullah AH, Gani A (2013) Imperialist competitive algorithm combined with refined high-order weighted fuzzy time series (RHWFTS-ICA) for short term load forecasting. Energy Convers Manag. https://doi.org/10.1016/j.enconman.2013.08.039
26. Lee WJ, Hong J (2015) A hybrid dynamic and fuzzy time series model for mid-term power load forecasting. Int J Electr Power Energy Syst. https://doi.org/10.1016/j.ijepes.2014.08.006
27. Sadaei HJ, de Lima e Silva PC, Guimarães FG, Lee MH (2019) Short-term load forecasting by using a combined method of convolutional neural networks and fuzzy time series. Energy. https://doi.org/10.1016/j.energy.2019.03.081
28. Wang T, Lai CS, Ng WWY et al (2021) Deep autoencoder with localized stochastic sensitivity for short-term load forecasting. Int J Electr Power Energy Syst. https://doi.org/10.1016/j.ijepes.2021.106954
29. Zhao W, Li T, Xu D, Wang Z (2022) A global forecasting method of heterogeneous household short-term load based on pre-trained autoencoder and deep-LSTM model. Ann Oper Res. https://doi.org/10.1007/s10479-022-05070-y
30. Berahmand K, Daneshfar F, Salehi ES et al (2024) Autoencoders and their applications in machine learning: a survey. Artif Intell Rev. https://doi.org/10.1007/s10462-023-10662-6
31. Berahmand K, Li Y, Xu Y (2023) DAC-HPP: deep attributed clustering with high-order proximity preserve. Neural Comput Appl. https://doi.org/10.1007/s00521-023-09052-4
32. Stefenon SF, Ribeiro MHDM, Nied A et al (2021) Hybrid wavelet stacking ensemble model for insulators contamination forecasting. IEEE Access. https://doi.org/10.1109/ACCESS.2021.3076410
33. Shakeel A, Chong D, Wang J (2023) District heating load forecasting with a hybrid model based on LightGBM and FB-prophet. J Clean Prod. https://doi.org/10.1016/j.jclepro.2023.137130
34. Li Y, Zhu N, Hou Y (2023) A novel hybrid model for building heat load forecasting based on multivariate Empirical modal decomposition. Build Environ. https://doi.org/10.1016/j.buildenv.2023.110317
35. Nti IK, Teimeh M, Nyarko-Boateng O, Adekoya AF (2020) Electricity load forecasting: a systematic review. J Electr Syst Inf Technol. https://doi.org/10.1186/s43067-020-00021-8
36. Candela Esclapez A, López García M, Valero Verdú S, Senabre Blanes C (2022) Automatic selection of temperature variables for short-term load forecasting. Sustain (Switzerland). https://doi.org/10.3390/su142013339
37. Akhtar S, Shahzad S, Zaheer A et al (2023) Short-term load forecasting models: a review of challenges, progress, and the road ahead. Energies (Basel). https://doi.org/10.3390/en16104060
38. Atanassov KT (1986) Intuitionistic fuzzy sets. Fuzzy Sets Syst. https://doi.org/10.1016/S0165-0114(86)80034-3
39. Gross G, Galiana FD (1987) Short-term load forecasting. Proc IEEE 75:1558–1573. https://doi.org/10.1109/PROC.1987.13927
40. Zadeh LA (1965) Fuzzy sets. Inf Control. https://doi.org/10.1016/S0019-9958(65)90241-X
41. Chaira T (2011) A novel intuitionistic fuzzy C means clustering algorithm and its application to medical images. Appl Soft Comput J 11(2):1711–1717
42. Demuth H, Raele MH (2009) Neural network toolbox user's guide for use with MATLAB. The MathWorks
43. Fahlman SE, Lebiere C (1990) The cascade-correlation learning architecture. In: Advances in neural information processing
44. Fahlman SE, Lebiere C (1997) The cascade-correlation learning architecture scott. In: Proceedings of NIPS
45. Alkhasawneh MS (2019) Hybrid cascade forward neural network with elman neural network for disease prediction. Arab J Sci Eng. https://doi.org/10.1007/s13369-019-03829-3

46. Deng S, Chen F, Wu D et al (2022) Quantitative combination load forecasting model based on forecasting error optimization. Comput Electr Eng. https://doi.org/10.1016/j.compeleceng.2022.108125
47. Gulay E, Sen M, Akgun OB (2024) Forecasting electricity production from various energy sources in Türkiye: a predictive analysis of time series, deep learning, and hybrid models. Energy. https://doi.org/10.1016/j.energy.2023.129566